

A RELATIVE MÖBIUS–BAR COMPLEX FOR POLYNOMIAL PRIME CORRELATIONS

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ABSTRACT. We construct an explicit divisor–bar model for polynomial prime correlations over $\mathbb{F}_q[T]$. The effective-divisor interval complex compresses on the squarefree locus to the sign local system. Its degree-weighted sum has alternating trace Λ ; after the necessary even regrading, an explicit Koszul differential identifies it globally with Sawin’s exterior-power factorization package. Geometrically this is the Euler-tail transform of the normalized Das–Howe bar. Marking the terminal-rank interval gives a flat, associative connection-corrected dg convolution, a resultant-supported collision triangle, and a chain-level Čech cube whose Grothendieck class is the Heath–Brown identity.

For the universal pair $(f, f + h)$ these objects give a relative complex whose trace is the von Mangoldt two-point correlation. Its squarefree-divisor expansion has an affine-bundle range whose Euler product yields the usual singular series, while the complement is isolated as a residual complex. This localizes, but does not formally remove, the parity problem.

For the ray modulus $D\infty^2$, with D squarefree of degree k , we construct the filtered Fourier object and prove an arbitrary-characteristic interior microlocal cutoff, simultaneous boundary normal forms, fixed-frequency perversity, and concentration in two adjacent degrees. The special-stalk polar term is bounded by $P_{n,k} = 2^{k-2} \binom{n+k-2}{2k-3}$ for $k \geq 2$ and by zero for $k = 1$. The true Lefschetz–Verdier local term at a wild fixed cycle retains its nonreduced graph–diagonal intersection length; this gives the characteristic-free generic rank $G_{n,k} = 2^{k-1} \binom{n+k-1}{2k-1}$ and the pointwise estimate

$$|S_\alpha(n; D, a)| \leq (G_{n,k} + \sqrt{q} P_{n,k}) q^{(n-k)/2}.$$

For $\deg D = 1$ the prime sum is exactly hyper-Kloosterman. The resulting fixed- q bounded-additive-sieve range crosses the $1/2$ divisor-degree threshold for $q > 266$. The coherent Type I/II comparison and the Goldbach parity column remain separate open problems.

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1. PURPOSE AND RELATION TO EXISTING WORK

The polynomial identity

$$(1.1) \quad \Lambda(f) = - \sum_{D|f} \mu(D) \deg D$$

is normally used as a numerical convolution. Our goal is to lift (1.1) to an object of an ℓ -adic derived category and then apply the six-functor formalism before taking traces.

There are three levels of content, which should not be conflated.

- (1) The equality between a poset Möbius function and the reduced Euler characteristic of an order complex is classical; see Rota [17] and Björner [4]. Its functorial categorification for poschemes, including the effective zero-cycle calculation used below, is due to Das–Howe [7].
- (2) The use of local systems and the Grothendieck–Lefschetz formula to study factorization statistics of finite-field polynomials is also established; see Church–Ellenberg–Farb [5], Hyde [13], and Petersen–Tosteson [16]. More directly, Sawin defines factorization functions in [21, Definition 1.6]; the Möbius and von Mangoldt identities appear in [22, Lemmas 3.5 and 3.6], with the latter restated as [21, Lemma 8.13]. The von Mangoldt statement is an alternating character identity; it does not by itself specify a non-split derived prime complex.
- (3) A method explicitly called the *homological sieve*, based on poschemes, bar complexes, truncation, and Frobenius traces, was developed by Das–Lehmann–Tanimoto–Tosteson [6]; see also the survey [24]. Their arithmetic applications concern spaces of rational curves and Manin’s conjecture.

Thus neither the name “homological sieve” nor bar-complex inclusion–exclusion is new here. The construction below is a specialized interface between those ideas and polynomial prime correlations. The proved statements in this note are a convenient synthesis of existing cohomological machinery with elementary stable-range calculations. The degree-weighted Koszul comparison in [theorem 4.5](#) supplies a first chain-level bridge between the two languages, and [theorem 4.11](#) identifies that bridge with the Euler-tail of the genuine terminal-rank-filtered poscheme bar. The core filtered antisymmetrization is [7, Theorem 6.5.4]; the specialization made here is that integrating all filtration tails produces exactly the degree weight in von Mangoldt inversion. The next new-content step is the marked rank-interval construction in [theorem 4.18](#): it upgrades the graded product rule to a connection-corrected dg statement and separates the common-factor defect in [proposition 4.21](#). These are formal-algebraic consequences of the filtered bar machinery, not cancellation estimates. The next arithmetic threshold is deliberately higher: identify the cubical bar resolution with genuine Type I/II coefficient objects, or deduce a new power-saving estimate from the residual weight statement in [research Target 8.1](#). Throughout, $|F| := q^{\deg F}$ for every nonzero polynomial F .

2. THE DIVISOR BAR COMPLEX

Let P be a finite poset and let $x < y$ in P . Write $\Delta(x, y)$ for the order complex of the open interval

$$(x, y) = \{z \in P : x < z < y\}.$$

We use augmented reduced cochains, so that the empty complex has one copy of the coefficient field in degree -1 .

Proposition 2.1 (Rota). *For any finite poset P ,*

$$\mu_P(x, y) = \tilde{\chi}(\Delta(x, y)).$$

Proof. The reduced Euler characteristic is the alternating chain sum

$$-1 + \sum_{r \geq 1} (-1)^{r-1} \#\{x < z_1 < \cdots < z_r < y\}.$$

Grouping chains according to their last element shows that this function of (x, y) satisfies

$$\sum_{x \leq z \leq y} \tilde{\chi}(\Delta(x, z)) = 0 \quad (x < y),$$

with value 1 on the diagonal. These are the defining recursions for the incidence-algebra inverse of the zeta function. $\square$

Let $F \in \mathbb{F}_q[T]$ be monic and let $\text{Div}(F)$ be its divisor poset. Then

$$(2.1) \quad \mu(F) = \mu_{\text{Div}(F)}(1, F).$$

If F is squarefree with r distinct irreducible factors, the interval $(1, F)$ is the poset of nonempty proper subsets of an r -element set. Its order complex is the barycentric subdivision of the boundary of an $(r - 1)$ -simplex. In particular its reduced homology is one-dimensional, concentrated in degree $r - 2$, and its reduced Euler characteristic is $(-1)^{r-2} = (-1)^r = \mu(F)$. The point of the next section is to globalize this orientation line while the polynomial varies.

3. BAR COMPRESSION AND THE MÖBIUS LOCAL SYSTEM

Let

$$\mathbf{M}_a = \text{Spec } \mathbb{F}_q[c_0, \dots, c_{a-1}]$$

be the affine space of monic degree- a polynomials, and let $\mathbf{U}_a \subset \mathbf{M}_a$ be the discriminant complement. The ordered-root cover

$$\tilde{\mathbf{U}}_a \longrightarrow \mathbf{U}_a$$

is a finite étale S_a -torsor. Fix a prime $\ell \nmid q$ and let $\mathcal{S}_a$ be the rank-one lisse $\overline{\mathbb{Q}}_\ell$ -sheaf associated with the sign representation of S_a .

For a constructible complex K at an $\mathbb{F}_q$ -point, put

$$\text{Tr}^{\text{alt}}(\text{Frob}_q, K) = \sum_i (-1)^i \text{Tr}(\text{Frob}_q, H^i(K)).$$

Let $j_a : \mathbf{U}_a \hookrightarrow \mathbf{M}_a$ be the open immersion. Over $\mathbf{M}_a$, let $\mathbf{l}_a$ be the relative poscheme whose fiber over the effective zero-cycle D is the open interval

$$(0, D) = \{E : 0 < E < D\}$$

of proper nonzero effective subcycles. Write $\varepsilon : N_\bullet \mathbf{l}_a \rightarrow \mathbf{M}_a$ for the augmentation of its nerve. Here $R\varepsilon_*$ means the normalized totalization of the simplicial pushforward. Working in the stable ℓ -adic derived ∞ -category, define the reduced interval object functorially by

$$(3.1) \quad \tilde{\mathcal{B}}_a := \text{cofib} \left(\overline{\mathbb{Q}}_{\ell \mathbf{M}_a} \longrightarrow R\varepsilon_* \overline{\mathbb{Q}}_{\ell N_\bullet \mathbf{l}_a} \right).$$

Theorem 3.1 (Effective-divisor bar compression). *There is a canonical isomorphism*

$$(3.2) \quad \tilde{\mathcal{B}}_a \simeq j_{a,!} \mathcal{S}_a[2-a] \quad \text{in } D_c^b(\mathbf{M}_a, \overline{\mathbb{Q}}_\ell).$$

Thus the interval object vanishes on every non-squarefree divisor.

Proof. Apply [7, Proposition 6.3.2] to effective zero-cycles on $\mathbb{P}^1$, then base-change along the open immersion $\mathrm{Sym}^a \mathbb{A}^1 \hookrightarrow \mathrm{Sym}^a \mathbb{P}^1$. On the ordered-root cover of $\mathbf{U}_a$, the interval is the poset of nonempty proper subsets of $\{1, \dots, a\}$. Its nerve is the barycentric subdivision of the boundary of an $(a-1)$ -simplex; its only reduced cohomology is the orientation line in degree $a-2$, and S_a acts on that line through the sign representation. This gives $\mathcal{S}_a[2-a]$ on $\mathbf{U}_a$. The cited result also proves extension by zero across the non-squarefree locus. Concretely, if a geometric divisor has multiplicity vector $(e_1, \dots, e_r)$ with some $e_j \geq 2$, its proper divisor interval is the proper part of $\prod_i [0, e_i]$. The closure map $v \mapsto v \vee u_j$, where u_j has a single 1 in the j th coordinate, has image with minimum u_j and never reaches the deleted top element. Its nerve is therefore contractible. For $a=1$ the interval is empty, and the cofiber convention gives $\overline{\mathbb{Q}}_\ell[1]$, as required. $\square$

Definition 3.2 (Trace-normalized Möbius object). For $a \geq 1$, define

$$\mathcal{M}_a := \tilde{\mathcal{B}}_a \simeq j_{a,!} \mathcal{S}_a[2-a].$$

Remark 3.3 (The shift audit). Conventions matter before one takes traces. In the normalization of [6, §7], the canonical interval Möbius object μ_a satisfies

$$\mu_a[2] = \tilde{\mathcal{B}}_a, \quad \mu_a \simeq j_{a,!} \mathcal{S}_a[-a].$$

Our $\mathcal{M}_a$ is therefore $\mu_a[2]$. The even shift leaves its alternating trace unchanged. The perverse-normalized shift $j_{a,!} \mathcal{S}_a[a]$ has the same trace again, but differs from μ_a by the even shift $[2a]$. These objects must not be identified merely because their trace functions agree.

Proposition 3.4 (Pointwise Möbius trace). *For every $D \in \mathbf{M}_a(\mathbb{F}_q)$,*

$$\mathrm{Tr}^{\mathrm{alt}}(\mathrm{Frob}_D, (\mathcal{M}_a)_{\bar{D}}) = \mu(D).$$

Proof. If D is not squarefree, both sides vanish by [theorem 3.1](#). Suppose that D is squarefree and has r irreducible factors. Frobenius acts on its a geometric roots as a permutation with r cycles, hence

$$\mathrm{Tr}(\mathrm{Frob}_D, (\mathcal{S}_a)_{\bar{D}}) = \mathrm{sgn}(\mathrm{Frob}_D) = (-1)^{a-r}.$$

The shift $[2-a]$ contributes $(-1)^{a-2} = (-1)^a$ to the alternating trace. Their product is $(-1)^r = \mu(D)$. $\square$

Remark 3.5. [Theorem 3.1](#) is the first useful compression: an entire simplicial divisor resolution is replaced by one rank-one local system, without taking an Euler characteristic. It is nevertheless a standard instance of the Das–Howe effective-zero-cycle calculation, not a new cancellation estimate. Its useful algebraic content is stronger than the numerical statement $\mu(D) = 0$ for nonsquarefree D : the entire interval complex is zero there.

4. A DERIVED LIFT OF THE VON MANGOLDT FUNCTION

For $1 \leq a \leq d$, multiplication gives a morphism

$$m_{a,d} : \mathbf{M}_a \times \mathbf{M}_{d-a} \longrightarrow \mathbf{M}_d, \quad (D, N) \longmapsto DN.$$

Define the Möbius–degree object and the prime object on $\mathbf{M}_d$ by

$$(4.1) \quad \mathcal{A}_d := \bigoplus_{a=1}^d (Rm_{a,d,!} \mathrm{pr}_1^* \mathcal{M}_a)^{\oplus a},$$

$$(4.2) \quad \mathcal{P}_d := \mathcal{A}_d[1].$$

Theorem 4.1 (The von Mangoldt trace). *For every $f \in \mathbf{M}_d(\mathbb{F}_q)$,*

$$\mathrm{Tr}^{\mathrm{alt}}(\mathrm{Frob}_f, (\mathcal{A}_d)_{\bar{f}}) = -\Lambda(f), \quad \mathrm{Tr}^{\mathrm{alt}}(\mathrm{Frob}_f, (\mathcal{P}_d)_{\bar{f}}) = \Lambda(f).$$

Proof. The $\mathbb{F}_q$ -points of the fiber of $m_{a,d}$ over f are the monic divisors $D \mid f$ of degree a ; the quotient $N = f/D$ is then unique. The nonsquarefree points contribute zero. The Grothendieck–Lefschetz trace formula for this zero-dimensional fiber and [proposition 3.4](#) give

$$\mathrm{Tr}^{\mathrm{alt}}(\mathrm{Frob}_f, (\mathcal{A}_d)_{\bar{f}}) = \sum_{\substack{D \mid f \\ D \text{ squarefree}}} \mu(D) \deg D.$$

The restriction to squarefree divisors causes no loss because μ vanishes on all other divisors. Equation (1.1) gives the first identity, and the shift [1] gives the second. $\square$

Remark 4.2 (What has and has not happened). The object $\mathcal{P}_d$ is a genuine derived lift of Λ : it is functorial under pullback and can be tensor-multiplied before taking traces. However, [theorem 4.1](#) is still a split categorification of Möbius inversion. Arithmetic cancellation only enters after studying the cohomology of a pushforward containing $\mathcal{P}_d$.

4.1. A degree-weighted Koszul comparison. We first correct a possible ambiguity in the word “complex.” Let

$$q_d : \mathbb{A}^d \longrightarrow \mathbf{M}_d$$

send an ordered list of roots to the corresponding monic polynomial. For an S_d -representation R over $\overline{\mathbb{Q}}_\ell$, put

$$(4.3) \quad \mathbf{F}_d(R) := (q_{d,*} \overline{\mathbb{Q}}_\ell \otimes R)^{S_d}.$$

The map q_d is finite, and invariants are exact over $\overline{\mathbb{Q}}_\ell$; hence $\mathbf{F}_d$ is an exact functor from S_d -representations to constructible sheaves. Sawin’s identity [[22](#), Lemma 3.6] (also [[21](#), Lemma 8.13]) is the equality of trace functions

$$\Lambda = \sum_{i=0}^{d-1} (-1)^i F_{\wedge^i \mathrm{std}_d}.$$

Thus

$$(4.4) \quad \mathcal{L}_d^{\mathrm{std}} := \bigoplus_{i=0}^{d-1} \mathbf{F}_d(\wedge^i \mathrm{std}_d)[i]$$

is a convenient derived lift of that identity, but the direct sum and its shifts are a choice made here; Sawin’s lemma does not assert a non-split differential between these summands.

Let $V_d = \overline{\mathbb{Q}}_\ell^d$ be the permutation representation, let

$$\epsilon : V_d \longrightarrow \overline{\mathbb{Q}}_\ell, \quad (x_1, \dots, x_d) \longmapsto \sum_j x_j, \quad W_d := \ker \epsilon = \mathrm{std}_d,$$

and set $u = d^{-1} \sum_j e_j$, so that $V_d = \overline{\mathbb{Q}}_\ell u \oplus W_d$ and $\epsilon(u) = 1$. For $1 \leq a \leq d$, write

$$(4.5) \quad I_a := \mathrm{Ind}_{S_a \times S_{d-a}}^{S_d} (\mathrm{sgn}_a \boxtimes \mathbf{1}) \simeq \wedge^a V_d.$$

Lemma 4.3 (Divisor layers as factorization sheaves). *There are canonical isomorphisms*

$$Rm_{a,d,!}(j_{a,!} \mathcal{S}_a \boxtimes \overline{\mathbb{Q}}_\ell) \simeq \mathbf{F}_d(I_a)$$

and consequently

$$Rm_{a,d,!} \mathrm{pr}_1^* \mathcal{M}_a \simeq \mathbf{F}_d(I_a)[2-a].$$

They hold on all of $\mathbf{M}_d$, including the discriminant.

Proof. All maps in question are finite, so $!$ and $*$ agree. On the ordered root cover, passing from $S_a \times S_{d-a}$ to S_d is induction. Frobenius reciprocity identifies its invariant pushforward with $\mathbf{F}_d(I_a)$. The sign factor vanishes at a repeated root because the corresponding stabilizer contains a transposition; hence its descent is exactly $j_{a,!} \mathcal{S}_a$. The second assertion follows from [theorem 3.1](#). $\square$

The shifts in this lemma matter. Define a second lift of Sawin's alternating formula by

$$(4.6) \quad \mathcal{L}_d^{\text{bar}} := \bigoplus_{i=0}^{d-1} F_d(\wedge^i W_d)[2-i].$$

The shifts in (4.4) and (4.6) have the same parity, so both objects have trace Λ , but they are not generally quasi-isomorphic.

Proposition 4.4 (The shift obstruction). *For $d \geq 2$ and $f = (T - c)^d$,*

$$(\mathcal{P}_d)_{\bar{f}} \simeq \overline{\mathbb{Q}}_\ell[2], \quad (\mathcal{L}_d^{\text{std}})_{\bar{f}} \simeq \overline{\mathbb{Q}}_\ell[0], \quad (\mathcal{L}_d^{\text{bar}})_{\bar{f}} \simeq \overline{\mathbb{Q}}_\ell[2].$$

In particular, no finite filtered complex whose associated graded is literally the divisor-graded object $\mathcal{P}_d$ can have $\mathcal{L}_d^{\text{std}}$ as its total object.

Proof. At f a squarefree divisor of positive degree can only have degree one. Thus the stalk of (4.2) has only its $a = 1$ layer, which is $\overline{\mathbb{Q}}_\ell[2]$. The stabilizer of an ordered root list is S_d . Consequently

$$(F_d(\wedge^i W_d))_{\bar{f}} = (\wedge^i W_d)^{S_d},$$

which is zero for $i > 0$ and is $\overline{\mathbb{Q}}_\ell$ for $i = 0$. This proves all three identities. At this stalk a filtration with associated graded $\mathcal{P}_d$ has only one nonzero graded piece, so no later differential can change its cohomological degree. $\square$

The obstruction has a precise repair in either of two normalizations. The first retains the original divisor shifts and changes the exterior package; the second retains (4.4) and applies an even cohomological regrading to the divisor layers.

Theorem 4.5 (Degree-weighted Koszul comparison). *Let $E_a = \overline{\mathbb{Q}}_\ell[z]_{<a}$.*

(1) *Put*

$$(\mathcal{K}_d^{\text{bar}})^{a-3} = F_d(I_a) \otimes E_a$$

and define

$$\delta^{a-3} = F_d(u \wedge -) \otimes (E_a \hookrightarrow E_{a+1}).$$

Then $\delta^2 = 0$. The decreasing divisor-degree filtration has

$$\text{gr}_a \mathcal{K}_d^{\text{bar}} \simeq F_d(I_a)^{\oplus a}[3-a],$$

so its associated graded is exactly $\mathcal{P}_d$, and there is a quasi-isomorphism

$$\mathcal{K}_d^{\text{bar}} \xrightarrow{\sim} \mathcal{L}_d^{\text{bar}}.$$

(2) *Put $U_a = \overline{\mathbb{Q}}_\ell[t]/(t^a)$, $U_0 = 0$, and let $p_a : U_a \rightarrow U_{a-1}$ be truncation. Define*

$$(\mathcal{K}_d^{\text{std}})^{1-a} = F_d(I_a) \otimes U_a, \quad \partial^{1-a} = F_d(\iota_\epsilon) \otimes p_a,$$

where ι_ϵ is contraction by ϵ . Then $\partial^2 = 0$ and

$$\mathcal{K}_d^{\text{std}} \xrightarrow{\sim} \mathcal{L}_d^{\text{std}}.$$

Its divisor-degree graded pieces are

$$\text{gr}_a \mathcal{K}_d^{\text{std}} \simeq F_d(I_a)^{\oplus a}[a-1].$$

Thus they are obtained from the a th graded piece of $\mathcal{P}_d$ by the necessary even shift $[2a-4]$.

Both comparisons are global on $\mathbf{M}_d$, not merely comparisons on the squarefree locus or in K_0 .

Proof. It suffices first to work with S_d -representations, because F_d is exact.

For part (1), decompose E_a into its monomial lines. For fixed r , the line $\overline{\mathbb{Q}}_\ell z^r$ occurs exactly for $a \geq r+1$, and gives the truncated co-Koszul complex

$$\wedge^{r+1} V_d \xrightarrow{u \wedge -} \wedge^{r+2} V_d \longrightarrow \cdots \longrightarrow \wedge^d V_d.$$

Using $V_d = \overline{\mathbb{Q}}_\ell u \oplus W_d$, wedge multiplication kills $u \wedge \wedge^\bullet W_d$ and maps $\wedge^\bullet W_d$ isomorphically onto the corresponding u -summand in the next term. The displayed tail therefore has only one cohomology representation, namely $\wedge^r W_d$, represented at its first term by

$$\eta \longmapsto u \wedge \eta \otimes z^r.$$

That first term lies in degree $r - 2$, so the resulting derived summand is $\wedge^r W_d[2 - r]$. Summing over r proves the first quasi-isomorphism. The associated-graded formula follows from $\dim E_a = a$ and [lemma 4.3](#).

For part (2), the contraction satisfies $\iota_\epsilon^2 = 0$. The monomial line $\overline{\mathbb{Q}}_\ell t^r$ occurs exactly in U_a for $a \geq r + 1$ and gives the opposite truncated Koszul tail

$$\wedge^d V_d \longrightarrow \cdots \longrightarrow \wedge^{r+1} V_d.$$

The decomposition

$$\wedge^a V_d = \wedge^a W_d \oplus u \wedge \wedge^{a-1} W_d$$

shows that its sole cohomology is $\wedge^r W_d$ in degree $-r$: contraction maps the second summand isomorphically to the first summand one step lower. Hence this tail is quasi-isomorphic to $\wedge^r W_d[r]$. Summing over r proves the second comparison. Finally, the old and new shifts differ by

$$(a - 1) - (3 - a) = 2a - 4.$$

□

Corollary 4.6 (The divisor-degree spectral sequence). *The filtration in [theorem 4.5\(1\)](#) gives a convergent finite spectral sequence whose first page consists of the original split divisor layers*

$$E_1^{a,*} = F_d(I_a)^{\oplus a}[3 - a] \implies \mathcal{L}_d^{\text{bar}}.$$

Its first differential is induced by $u \wedge - \otimes (E_a \hookrightarrow E_{a+1})$. For any separated finite-type map $\rho : \mathbf{M}_d \rightarrow Y$, applying $R\rho_!$ gives

$$R\rho_! F_d(I_a)^{\oplus a}[3 - a] \implies R\rho_! \mathcal{L}_d^{\text{bar}}.$$

Proof. This is the spectral sequence of the finite decreasing filtration. Derived compactly supported pushforward preserves its finite totalization, differentials, and quasi-isomorphisms. □

Remark 4.7 (Rees form). Introduce a parameter $\hbar$ and replace δ by $\hbar\delta$. The fiber at $\hbar = 0$ is the split object $\mathcal{P}_d$, while the fiber at $\hbar = 1$ is $\mathcal{K}_d^{\text{bar}}$ and retracts to $\mathcal{L}_d^{\text{bar}}$. This is a genuine non-split dg/Rees enhancement, but it is a degeneration rather than a family of fiberwise quasi-isomorphisms: its cohomology is allowed to jump at $\hbar = 0$.

Remark 4.8 (The Euler derivative is visible on chains). In the representation ring,

$$\lambda_{-v}(V_d) = \sum_{a=0}^d (-v)^a [\wedge^a V_d] = (1 - v) \lambda_{-v}(W_d),$$

and therefore

$$-v \frac{d}{dv} \lambda_{-v}(V_d) \Big|_{v=1} = \lambda_{-1}(W_d).$$

The left side contains a copies of $\wedge^a V_d$. In [theorem 4.5](#), that multiplicity is realized by the a monomial lines of E_a or U_a , and each monomial selects one truncated Koszul tail. Thus the theorem is a chain-level categorification of the Euler derivative, rather than a lift chosen only after checking the K_0 identity.

Remark 4.9 (Why a Grothendieck-class inverse is insufficient). The numerical identity $\mu * 1 = \delta_1$ lifts to an equality in K_0 , but not to a direct-sum identity in a derived category. Already over the divisor lattice of one prime P , the split stalk is $\overline{\mathbb{Q}}_\ell \oplus \overline{\mathbb{Q}}_\ell[1]$: its class and alternating trace vanish, while the object is not zero. The Koszul differential joining the two summands is what makes the totalization contractible. The two differentials in [theorem 4.5](#) perform exactly this cancellation across successive divisor degrees.

4.2. The geometric Euler-tail of the zero-cycle bar. Let $X = \mathbf{M}_d$, and let

$$\pi_d : Z_d := \operatorname{Spec}_X(\mathcal{O}_X[T]/(T^d + c_{d-1}T^{d-1} + \cdots + c_0)) \longrightarrow X$$

be the universal root scheme. It is finite flat of degree d . Put

$$\Gamma_d^{[1,d]} := \prod_{a=1}^d \Gamma_X^a(Z_d), \quad \varepsilon_\Gamma : N_\bullet \Gamma_d^{[1,d]} \longrightarrow X.$$

Let $C_{d,\Gamma}^\bullet$ be the normalized skeletal cochain complex of the simplicial pushforward $R\varepsilon_{\Gamma,*}\overline{\mathbb{Q}}_\ell$. In the monoidal coordinates of [7, §6.2],

$$(4.7) \quad C_{d,\Gamma}^p = \bigoplus_{\substack{\underline{a}=(a_0,\dots,a_p) \in \mathbb{Z}_{\geq 0}^{p+1} \\ |\underline{a}| \leq d}} (\pi_{\underline{a}})_* \overline{\mathbb{Q}}_\ell, \quad d = \sum_{i=0}^{p+1} (-1)^i \delta_i^*.$$

Here $\pi_{\underline{a}}$ is the finite projection from the product of relative cycle spaces indexed by $\underline{a}$; for $i < p+1$ the face δ_i merges adjacent increments, and δ_{p+1} forgets the last increment. Equip this complex with the decreasing terminal-rank filtration

$$(4.8) \quad F^r C_{d,\Gamma}^p = \bigoplus_{\substack{\underline{a} > 0, |\underline{a}| \leq d \\ |\underline{a}| > r}} (\pi_{\underline{a}})_* \overline{\mathbb{Q}}_\ell, \quad 0 \leq r \leq d.$$

There is also a literal divisor version. Let

$$\mathcal{D}_d^a := \mathbf{M}_a \times \mathbf{M}_{d-a} \xrightarrow{m_{a,d}} \mathbf{M}_d, \quad \mathcal{D}_d := \prod_{a=1}^d \mathcal{D}_d^a,$$

and order (D, N) below (D', N') when $D' = DE$ and $N = EN'$. Thus the geometric fiber over f is the poset of nonzero monic divisors of f . Let $C_{d,\operatorname{div}}^\bullet$ be the normalized cochain complex of its nerve, with the analogous terminal-degree filtration.

Lemma 4.10 (Divisor and zero-cycle bars). *The root-cycle map from $\mathcal{D}_d$ to $\Gamma_d^{[1,d]}$ induces a filtered quasi-isomorphism*

$$(C_{d,\operatorname{div}}^\bullet, F) \xrightarrow{\sim} (C_{d,\Gamma}^\bullet, F).$$

Proof. It is enough, by proper base change, to check the full nerve and every rank truncation on geometric fibers. If f has distinct roots $x_1, \dots, x_s$ with multiplicities $m_1, \dots, m_s$, the divisor fiber is

$$\left(\prod_{i=1}^s [0, m_i] \right) \setminus \{0\}.$$

The geometric points of the degree- $\leq d$ zero-cycle fiber are the nonzero vectors $(e_1, \dots, e_s)$ with $\sum_i e_i \leq d$. In both posets the support map

$$(e_i)_i \longmapsto (\min(e_i, 1))_i$$

is a falling retraction onto the same Boolean poset of nonempty subsets of $\{x_1, \dots, x_s\}$. It preserves every upper bound on total degree. Hence the map of full nerves and the map of each rank truncation are homotopy equivalences. Taking the successive cofibers gives the assertion for F^r and therefore for the filtered complexes. $\square$

For a complex C with the finite filtration $F^0 C \supset \cdots \supset F^d C = 0$, define its *Euler-tail* by

$$(4.9) \quad \Theta(C, F) := \bigoplus_{r=0}^{d-1} F^r C, \quad G^s \Theta(C, F) := \bigoplus_{r=0}^{d-1} F^{\max(r,s)} C.$$

This second filtration satisfies

$$(4.10) \quad \operatorname{Gr}_G^{a-1} \Theta(C, F) \simeq (\operatorname{Gr}_F^{a-1} C)^{\oplus a}.$$

Theorem 4.11 (Geometric Euler-tail realization). *With $\overline{\mathbb{Q}}_\ell$ -coefficients there are global filtered quasi-isomorphisms*

$$\Theta(C_{d,\text{div}}, F)[2] \xrightarrow{\sim} \Theta(C_{d,\Gamma}, F)[2] \xrightarrow{\sim} \mathcal{K}_d^{\text{bar}}.$$

For $1 \leq a \leq d$, the terminal-degree- a graded part is

$$\text{Gr}_G^{a-1} \Theta(C_{d,\Gamma}, F)[2] \simeq F_d \left(\text{Ind}_{S_a \times S_{d-a}}^{S_d} (\text{sgn}_a \boxtimes \mathbf{1}) \right)^{\oplus a} [3-a].$$

Consequently the G -associated graded is $\mathcal{P}_d$, while the total object is $\mathcal{L}_d^{\text{bar}}$. These comparisons commute with arbitrary base change and remain quasi-isomorphisms after any subsequent separated finite-type pushforward $R\rho_!$.

Proof. The first arrow is [lemma 4.10](#). For the second, apply [7, Theorem 6.5.4] with $S = X = \mathbf{M}_d$, $Z = Z_d$, and $K = \overline{\mathbb{Q}}_\ell$, and then quotient by the terminal-rank- $> d$ part. The hypotheses hold because Z_d/X is finite, hence projective, and $X \rightarrow S = X$ is proper. The resulting antisymmetrization is a filtered quasi-isomorphism

$$(C_{d,\Gamma}^\bullet, F) \xrightarrow{\sim} (B_d^\bullet, F), \quad B_d^{a-1} = (Z_d^{\times a} \rightarrow X)_* \overline{\mathbb{Q}}_\ell[\text{sgn}_a].$$

The factorization-sheaf description and Frobenius reciprocity identify

$$B_d^{a-1} \simeq F_d \left(\text{Ind}_{S_a \times S_{d-a}}^{S_d} (\text{sgn}_a \boxtimes \mathbf{1}) \right) \simeq F_d(\wedge^a V_d).$$

In the oriented-subset basis the alternating face differential is exactly

$$F_d(\omega_d \wedge -), \quad \omega_d = e_1 + \cdots + e_d.$$

Indeed, the coefficient of an oriented $(a+1)$ -subset is the Laplace expansion of the wedge with ω_d .

Since the filtration is finite, a filtered quasi-isomorphism induces a quasi-isomorphism on every F^r . Applying $\Theta = \bigoplus_r F^r$ therefore preserves the comparison. If $E_a = \overline{\mathbb{Q}}_\ell[z]_{<a}$ and z^r labels the copy contributed by F^r , then

$$\Theta(B_d, F)^{a-1} = F_d(\wedge^a V_d) \otimes E_a,$$

and its differential is

$$F_d(\omega_d \wedge -) \otimes (E_a \hookrightarrow E_{a+1}).$$

Writing $u = d^{-1}\omega_d$, multiplication of the degree- $(a-1)$ term by d^{1-a} changes this to the differential used in $\mathcal{K}_d^{\text{bar}}$. The global shift [2] places that term in cohomological degree $a-3$. Equation (4.10), together with [7, Theorem 4.0.2 and Proposition 6.3.2], gives the stated graded parts. The total-object assertion is [theorem 4.5\(1\)](#). Finally, base change and $R\rho_!$ preserve the finite sums, cofiber sequences, and quasi-isomorphisms used above. $\square$

Remark 4.12 (What the theorem does not say). The theorem does not identify $\mathcal{K}_d^{\text{bar}}$ with the unweighted full bar. The geometric fibers of either full poscheme bar retract to a nonempty Boolean poset with a maximum; hence

$$R\varepsilon_{\Gamma,*} \overline{\mathbb{Q}}_\ell \simeq \overline{\mathbb{Q}}_{\ell X}$$

and the reduced full bar vanishes. The prime complex is instead $\Theta(C_{d,\Gamma}, F)[2]$, the direct sum of all terminal-rank tails. Thus “Euler-tail of the filtered full bar” is literal, while “the full bar itself” would be false.

Remark 4.13 (Standard shifts by Verdier duality). Ordinary Deligne décalage changes a filtration, not its total object, so it cannot turn $\mathcal{K}_d^{\text{bar}}$ into $\mathcal{K}_d^{\text{std}}$. There is instead a geometric dual description. Since $X = \mathbf{M}_d$ is smooth of dimension d and $q_d : \mathbb{A}^d \rightarrow X$ is finite,

$$\mathbb{D}_X F_d(R) \simeq F_d(R^\vee)[2d](d).$$

After the standard self-duality of V_d ,

$$\mathcal{K}_d^{\text{std}} \simeq \mathbb{D}_X(\Theta(C_{d,\Gamma}, F))[-2d](-d).$$

Duality reverses the cochain degree, changes $\omega_d \wedge -$ into contraction by ϵ , and changes $E_a \hookrightarrow E_{a+1}$ into $U_{a+1} \twoheadrightarrow U_a$. Under a later map ρ , the appropriate compatibility is relative duality $\mathbb{D}R\rho_! \simeq R\rho_*\mathbb{D}$.

Remark 4.14 (Rational coefficients). The splitting in the antisymmetrization theorem uses division by $a!$. The result above is therefore a $\overline{\mathbb{Q}}_\ell$ -coefficient statement. It does not automatically give a $\mathbb{Z}_\ell$ -model when $\ell \mid a!$; an integral refinement would require derived sign coinvariants or a divided-power replacement.

Proposition 4.15 (The graded product rule and its dg obstruction). *There is an ordered decomposition*

$$E_{a+b} = E_a \oplus z^a E_b, \quad (f, g) \mapsto f + z^a g.$$

It is associative in (a, b) and gives the Euler-weighted associated-graded construction the usual Leibniz rule under convolution. However, this decomposition is not preserved by the differential of $\mathcal{K}^{\text{bar}}$. More precisely, if $\iota_n : E_n \hookrightarrow E_{n+1}$ is the standard inclusion, then under the two adjacent decompositions the component that increases the first degree is

$$(f, g) \mapsto \left(f + g(0)z^a, \frac{g - g(0)}{z} \right).$$

Thus there is an off-diagonal connection term.

Proof. The vector-space decomposition and its associativity follow by cutting a polynomial of degree $< a + b$ before the coefficient of z^a . This realizes the identity $a + b = a + b$ on the two ordered summands and hence the Leibniz rule on the G -associated graded and in K_0 . Re-expanding

$$f + z^a g = (f + g(0)z^a) + z^{a+1} \frac{g - g(0)}{z}$$

after applying ι_{a+b} proves the last formula. $\square$

4.3. The marked rank-interval bar and its connection. The coefficient space E_a does not parametrize a chosen root. Its a basis vectors parametrize the unit intervals in a terminal-rank interval of length a . This observation gives a geometric marking before antisymmetrization.

Definition 4.16 (Marked rank-interval nerve). Let $\mathcal{P}/X$ be a finite ranked poscheme with ranks in $\{1, \dots, N\}$, and let $\mathcal{P}_{\leq r}$ be its lower-rank truncation. Write $(-)_+$ for adjoining a disjoint base point. In pointed simplicial sheaves put

$$\mathfrak{M}(\mathcal{P}) := \bigvee_{r=0}^{N-1} \text{cofib}((N_\bullet \mathcal{P}_{\leq r})_+ \longrightarrow (N_\bullet \mathcal{P})_+).$$

A non-base-point simplex is a chain $x_0 \leq \dots \leq x_p$ together with $0 \leq r < \text{rk}(x_p)$. Thus r marks one unit interval in the rank range of the terminal vertex. The last face becomes the base point precisely when it lowers the terminal rank past r .

Lemma 4.17 (The mark realizes the Euler-tail). *Reduced normalized pushforward cochains give a functorial identification*

$$\tilde{C}^\bullet(\mathfrak{M}(\mathcal{P}), \overline{\mathbb{Q}}_\ell) \cong \bigoplus_{r=0}^{N-1} F^r C^\bullet(\mathcal{P}) = \Theta(C^\bullet(\mathcal{P}), F).$$

The divisor-to-zero-cycle comparison of lemma 4.10 remains a quasi-isomorphism after this marking, also for a finite disjoint union of coloured root schemes.

Proof. Reduced cochains on $N_\bullet \mathcal{P}/N_\bullet \mathcal{P}_{\leq r}$ are relative cochains for the lower truncation, hence exactly the summands with terminal rank $> r$. Taking the finite wedge over r gives the first claim. On geometric fibers, the support retraction used in lemma 4.10 preserves every lower rank truncation. It is therefore an equivalence on every displayed cofiber, colour by colour. $\square$

We now give the strict chain model. Put $E_0 = 0$, and for $a, b \geq 0$ write

$$P_{a,b} := E_a^R \oplus E_b^B, \quad \phi_{a,b}(f, g) := f + z^a g.$$

Thus $\phi_{a,b} : P_{a,b} \xrightarrow{\sim} E_{a+b}$. Let

$$\epsilon_b(g) = g(0), \quad S_b(g) = \frac{g - g(0)}{z}, \quad \beta_a(c) = cz^a.$$

The red and blue transport maps are

$$(4.11) \quad H_{a,b}(f, g) = \left(f + g(0)z^a, \frac{g - g(0)}{z} \right),$$

$$(4.12) \quad V_{a,b}(f, g) = (f, \iota_b g).$$

They satisfy

$$(4.13) \quad \phi_{a+1,b} H_{a,b} = \iota_{a+b} \phi_{a,b}, \quad \phi_{a,b+1} V_{a,b} = \iota_{a+b} \phi_{a,b},$$

$$(4.14) \quad H_{a,b+1} V_{a,b} = V_{a+1,b} H_{a,b}.$$

The last equation is the flatness of the discrete connection.

Let $F_{m,n}(R \boxtimes S) := F_m(R) \boxtimes F_n(S)$ on $\mathbf{M}_m \times \mathbf{M}_n$. With $\omega_m = \sum_{i=1}^m e_i$ and $\omega_n = \sum_{j=1}^n e_j$, define, for $1 \leq k \leq m+n$,

$$(4.15) \quad (\mathcal{K}_{m,n}^{\text{col}})^{k-3} := \bigoplus_{\substack{a+b=k \\ 0 \leq a \leq m, 0 \leq b \leq n}} F_{m,n}(\wedge^a V_m \boxtimes \wedge^b V_n) \otimes P_{a,b}.$$

On the (a, b) summand its red and blue differential components are

$$(4.16) \quad d_R = F_{m,n}(\omega_m \wedge - \boxtimes 1) \otimes H_{a,b},$$

$$(4.17) \quad d_B = (-1)^a F_{m,n}(1 \boxtimes \omega_n \wedge -) \otimes V_{a,b}.$$

The total exterior vector is $\omega_{m,n} := \omega_m \oplus \omega_n$. Multiplication of the total exterior-degree- k term by $(m+n)^{1-k}$ conjugates wedge multiplication by $\omega_{m,n}$ to wedge multiplication by

$$u_{m+n} := (m+n)^{-1} \omega_{m,n}.$$

Thus this single total normalization agrees with that used in [theorem 4.5](#).

Theorem 4.18 (Marked Euler-tail and dg convolution connection). *With $\overline{\mathbb{Q}}_\ell$ -coefficients:*

(1) *Equations (4.16)–(4.17) define a differential. Let*

$$(\mathcal{B}_{m,n}^\Theta)^{k-3} := \bigoplus_{a+b=k} F_{m,n}(\wedge^a V_m \boxtimes \wedge^b V_n) \otimes E_k,$$

with differential induced by $\omega_{m,n} \wedge - \otimes \iota_k$. This is the Euler-tail of the coloured Banerjee sign complex, shifted by [2]. Termwise, the maps

$$x \otimes y \otimes (f, g) \longmapsto (x \wedge y) \otimes (f + z^a g)$$

give a strict chain isomorphism

$$\mathcal{K}_{m,n}^{\text{col}} \xrightarrow{\sim} \mathcal{B}_{m,n}^\Theta.$$

(2) *Via [lemma 4.17](#) and Das–Howe antisymmetrization, there is a global quasi-isomorphism*

$$\mathcal{K}_{m,n}^{\text{col}} \xrightarrow{\sim} \tilde{C}^\bullet(\mathfrak{M}(\Gamma_{m,n}), \overline{\mathbb{Q}}_\ell)[2],$$

where $\Gamma_{m,n}$ is the effective-zero-cycle poscheme of the coloured disjoint union $Z_m \sqcup Z_n$. The assertion includes all within-colour collisions and every base point on the cross-colour resultant divisor.

- (3) The red coefficient summands form a subcomplex $\mathcal{K}_{m|n}^R$. The blue quotient $\mathcal{Q}_{m|n}^B$ has red transport S_b and blue transport ι_b . There is a distinguished triangle

$$\mathcal{K}_{m|n}^R \longrightarrow \mathcal{K}_{m,n}^{\text{col}} \longrightarrow \mathcal{Q}_{m|n}^B \xrightarrow{\nabla_{m|n}} \mathcal{K}_{m|n}^R[1],$$

where the connection is represented in bidegree (a, b) by

$$\nabla_{m|n} = F_{m,n}(\omega_m \wedge - \boxtimes 1) \otimes \beta_a \epsilon_b.$$

- (4) The construction is strictly associative for ordered iterated convolution and is homotopy coherent under permutations of the colours. It remains valid after finite multiplication pushforward and after any subsequent separated finite-type $R\rho_!$.

Proof. The pure red and pure blue squares vanish because exterior multiplication by ω_m or ω_n squares to zero. The mixed terms cancel by the Koszul sign and (4.14); hence $d^2 = 0$. The two identities in (4.13) prove the strict chain isomorphism in part (1). The Banerjee term of total terminal rank k lies in cochain degree $k - 1$; the global shift [2] places it in degree $k - 3$, exactly as in (4.15). Part (2) now follows from lemma 4.17 and the filtered quasi-isomorphism [7, Theorem 6.5.4], applied to every rank tail.

In the decomposition $P_{a,b} = E_a^R \oplus E_b^B$, the differential has the upper-triangular coefficient matrix

$$\begin{pmatrix} \iota_a & \beta_a \epsilon_b \\ 0 & S_b \end{pmatrix} \quad \text{in the red direction,} \quad \begin{pmatrix} 1 & 0 \\ 0 & \iota_b \end{pmatrix} \quad \text{in the blue direction.}$$

This gives the triangle and its connecting morphism. For ordered colours with degrees $a_1, \dots, a_c$, use

$$\bigoplus_{j=1}^c E_{a_j}^{(j)} \xrightarrow{\sim} E_{a_1+\dots+a_c}, \quad (f_j)_j \mapsto \sum_{j=1}^c z^{a_1+\dots+a_{j-1}} f_j.$$

All transport maps are conjugates of the same total inclusion $E_N \hookrightarrow E_{N+1}$, so every higher square commutes and the ordered construction is strictly associative. For a permutation σ of the colours, the comparison is the exterior block permutation, with its Koszul sign, tensored with $\phi_\sigma^{-1}\phi$ on the marked coefficient space. Since all ϕ 's identify that space with the same E_N , these comparisons satisfy the cocycle condition. The final functoriality follows because all sums are finite and multiplication is finite. $\square$

Corollary 4.19 (The coloured total and its trace). *For $m, n \geq 1$ there is a global quasi-isomorphism on $\mathbf{M}_m \times \mathbf{M}_n$*

$$\mathcal{K}_{m,n}^{\text{col}} \simeq \left(\mathcal{L}_m^{\text{bar}} \boxtimes \mathcal{L}_n^{\text{bar}} \right) [-2] \oplus \left(\mathcal{L}_m^{\text{bar}} \boxtimes \mathcal{L}_n^{\text{bar}} \right) [-3].$$

Consequently its pointwise alternating Frobenius trace is zero.

Proof. By theorems 4.5 and 4.18, including the total normalization above,

$$\mathcal{K}_{m,n}^{\text{col}} \simeq \bigoplus_{r=0}^{m+n-1} F_{m,n} \left(\wedge^r W_{m+n}|_{S_m \times S_n} \right) [2 - r].$$

Under $S_m \times S_n \subset S_{m+n}$,

$$W_{m+n}|_{S_m \times S_n} \cong W_m \oplus W_n \oplus \overline{\mathbb{Q}}_\ell \tau, \quad \tau = u_m - u_n,$$

and τ is invariant. Hence

$$\wedge^r (W_m \oplus W_n \oplus \overline{\mathbb{Q}}_\ell \tau) \cong \wedge^r (W_m \oplus W_n) \oplus \tau \wedge \wedge^{r-1} (W_m \oplus W_n).$$

Writing $s = r$ in the first summand and $s = r - 1$ in the second, their shifts are $[2 - s]$ and $[1 - s]$. Compared with the external product shift $[4 - s]$, these are exactly the global shifts $[-2]$ and $[-3]$. The two summands therefore have traces $\Lambda(f)\Lambda(g)$ and $-\Lambda(f)\Lambda(g)$. $\square$

Remark 4.20 (Why ordinary dg Leibniz is false). Let $A(V) = (\wedge^\bullet V, \omega \wedge -)$, including exterior degree zero, and let $D(V) = (\wedge^{\geq 1} V \otimes E_\bullet, \omega \wedge - \otimes \iota)$, without the global shifts. If $\epsilon(\omega) = \dim V$, then $(\dim V)^{-1} \iota_\epsilon$ contracts $A(V)$. Consequently the naive candidate

$$(D(V_m) \otimes A(V_n)) \oplus (A(V_m) \otimes D(V_n))$$

is contractible. In contrast, $D(V_m \oplus V_n)$ has exterior cohomology on the quotient by the total augmentation vector. For $m = n = 1$ its two cohomology groups both have dimension one. Thus the connection in [theorem 4.18](#) is genuine extension data, not a removable choice of basis.

4.4. The resultant collision triangle. Let $\mu_{m,n} : \mathbf{M}_m \times \mathbf{M}_n \rightarrow \mathbf{M}_{m+n}$ be multiplication. On the source there is a finite map of universal root schemes

$$i : Z_m \sqcup Z_n \longrightarrow \mu_{m,n}^* Z_{m+n},$$

induced on rings by

$$\mathcal{O}[T]/(fg) \longrightarrow \mathcal{O}[T]/(f) \times \mathcal{O}[T]/(g).$$

Proposition 4.21 (Collision triangle). *The unit of the finite map i induces a canonical filtered morphism*

$$\alpha_{m,n} : \mu_{m,n}^* \mathcal{K}_{m+n}^{\text{bar}} \longrightarrow \mathcal{K}_{m,n}^{\text{col}}.$$

It is a quasi-isomorphism on $U_{\text{res}} = \{\text{Res}(f, g) \neq 0\}$. Therefore

$$\mathcal{C}_{m,n}^{\text{coll}} := \text{fib}(\alpha_{m,n})$$

is supported on the resultant divisor and fits into

$$\mathcal{C}_{m,n}^{\text{coll}} \longrightarrow \mu_{m,n}^* \mathcal{K}_{m+n}^{\text{bar}} \longrightarrow \mathcal{K}_{m,n}^{\text{col}} \longrightarrow .$$

For every $(f, g) \in (\mathbf{M}_m \times \mathbf{M}_n)(\mathbb{F}_q)$,

$$\text{Tr}^{\text{alt}}(\text{Frob}_{(f,g)}, (\mathcal{C}_{m,n}^{\text{coll}})_{(f,g)}) = \Lambda(fg).$$

Proof. The map of root schemes induces maps on effective cycles preserving cycle degree. The unit $\overline{\mathbb{Q}}_\ell \rightarrow Ri_* \overline{\mathbb{Q}}_\ell$, followed by pushforward to the base, therefore gives a morphism of every relative rank tail in the asserted direction. Naturality of antisymmetrization gives $\alpha_{m,n}$. On U_{res} the Chinese remainder theorem makes i an isomorphism.

For a geometric point, write $m_\gamma = \text{ord}_\gamma(f)$ and $n_\gamma = \text{ord}_\gamma(g)$. After ordering the roots, the stabilizers are

$$H_{\text{un}} = \prod_{\gamma} S_{m_\gamma + n_\gamma}, \quad H_{\text{col}} = \prod_{\gamma} (S_{m_\gamma} \times S_{n_\gamma}) \subset H_{\text{un}}.$$

The stalk map is the degreewise inclusion

$$(\wedge^\bullet V_{m+n} \otimes E_\bullet)^{H_{\text{un}}} \hookrightarrow (\wedge^\bullet V_{m+n} \otimes E_\bullet)^{H_{\text{col}}}.$$

This also makes the support assertion explicit. Finally, [theorem 4.5](#) and [corollary 4.19](#) and additivity of traces in the displayed triangle give $\Lambda(fg) - 0$. $\square$

Remark 4.22 (Local colour-difference directions). Every distinct common root γ contributes the augmentation-zero vector

$$\delta_\gamma = n_\gamma \sum_{\text{red copies at } \gamma} e_i - m_\gamma \sum_{\text{blue copies at } \gamma} e_i.$$

These vectors complement $V_{m+n}^{H_{\text{un}}}$ inside $V_{m+n}^{H_{\text{col}}}$. Thus the collision object admits a natural filtration by the number of distinct roots of $\gcd(f, g)$. For $m = n = 1$, the map is an isomorphism off the diagonal; on the diagonal the fiber is $\overline{\mathbb{Q}}_\ell[0]$, whose trace is $\Lambda((T-a)^2) = 1$.

4.5. A cubical bar–Heath–Brown resolution. Work in the degree-graded derived category with convolution

$$(\mathcal{X} \star \mathcal{Y})_d = \bigoplus_{r+s=d} Rm_{r,s,!}(\mathcal{X}_r \boxtimes \mathcal{Y}_s),$$

and let $\mathbf{1}_\star$ be its degree-zero unit. Let $\mathcal{B}^{\text{aug}}$ be the augmented full divisor bar over all degrees, normalized as augmented reduced cochains shifted by $[-1]$. Thus its rank-zero free-cofactor term lies in cohomological degree zero, while its terminal-degree- a sign term is $F(I_a)[-a]$. In this normalization the canonical coaugmentation $\mathbf{1}_\star \rightarrow \mathcal{B}^{\text{aug}}$ is a quasi-isomorphism: in every positive total degree this is exactly the full-bar contraction of [remark 4.12](#). For $U \geq 0$ put

$$\mathcal{R}_U := F^U \mathcal{B}^{\text{aug}}, \quad \mathcal{Q}_U := \mathcal{B}^{\text{aug}} / \mathcal{R}_U.$$

Thus $\mathcal{R}_U$ contains terminal divisor degrees $> U$.

Proposition 4.23 (Cubical bar–Heath–Brown resolution). *For every degree-graded object $\mathcal{X}$ and every $J \geq 1$, tensoring J copies of the triangle $\mathcal{R}_U \rightarrow \mathbf{1}_\star \rightarrow \mathcal{Q}_U \rightarrow$ gives a canonical triangle*

$$\mathcal{X} \star \mathcal{R}_U^{\star J} \longrightarrow \mathcal{X} \longrightarrow \check{\mathcal{C}}_{J,U}(\mathcal{X}) \longrightarrow,$$

where

$$\check{\mathcal{C}}_{J,U}(\mathcal{X}) = \text{Tot} \left[(\mathcal{X} \star \mathcal{Q}_U)^{\oplus J} \longrightarrow (\mathcal{X} \star \mathcal{Q}_U^{\star 2})^{\oplus \binom{J}{2}} \longrightarrow \cdots \longrightarrow \mathcal{X} \star \mathcal{Q}_U^{\star J} \right]$$

has the alternating cubical face maps. If $\mathcal{X} = \mathcal{P}$ is the graded prime object, then

$$(\mathcal{P} \star \mathcal{R}_U^{\star J})_d = 0 \quad \text{for } d \leq J(U+1).$$

In that range the triangle is a finite chain-level resolution of $\mathcal{P}_d$.

Let $\mathcal{L}_d = \overline{\mathbb{Q}}_{\ell \mathbf{M}_d}$, $\mathcal{M}_0 = \overline{\mathbb{Q}}_{\ell \mathbf{M}_0}$, $\mathcal{M}_d = \mathcal{M}_d$ for $d > 0$, and let $\mathcal{L}_d = \overline{\mathbb{Q}}_{\ell \mathbf{M}_d}^{\oplus d}$. In K_0 ,

$$[\mathcal{Q}_U] = [\mathcal{L} \star \mathcal{M}_{\leq U}], \quad [\mathcal{P}] = [\mathcal{L} \star \mathcal{M}].$$

Consequently, in every total degree,

$$\begin{aligned} [\mathcal{P}_d] &= \sum_{j=1}^J (-1)^{j-1} \binom{J}{j} \left[\left(\mathcal{L} \star \mathcal{L}^{\star(j-1)} \star \mathcal{M}_{\leq U}^{\star j} \right)_d \right] \\ &\quad + \left[(\mathcal{P} \star \mathcal{R}_U^{\star J})_d \right]. \end{aligned}$$

For $d \leq J(U+1)$ the last term vanishes and this becomes precisely the function-field Heath–Brown identity

$$[\mathcal{P}_d] = \sum_{j=1}^J (-1)^{j-1} \binom{J}{j} \left[\left(\mathcal{L} \star \mathcal{L}^{\star(j-1)} \star \mathcal{M}_{\leq U}^{\star j} \right)_d \right].$$

Proof. The total fiber of the nonempty-subset Čech J -cube formed from $\mathbf{1}_\star \rightarrow \mathcal{Q}_U$ is $\mathcal{R}_U^{\star J}$, proving the triangle. Moreover $(\mathcal{R}_U)_r = 0$ for $r \leq U$, while $\mathcal{P}$ is supported in positive degrees. Hence its remainder has minimum total degree $1 + J(U+1)$.

Bar compression and the rank cutoff give the first K_0 identity. The second is the chain-class form of $\Lambda = (\deg) \star \mu$, equivalently the derivative of $\mathcal{L} \star \mathcal{M} = \mathbf{1}_\star$. Substituting these identities in the alternating class of the cube cancels one full $\mathcal{M} \star \mathcal{L}$ factor and gives the displayed formula, which is the degree version of [\[12, §2\]](#). $\square$

Remark 4.24 (Which analytic columns occur). The last identification in [proposition 4.23](#) is currently an identity in K_0 and of trace functions, not a quasi-isomorphism between each cubical vertex and the standard analytic coefficient object. Also, the proof of [\[23, Corollary 6.1\]](#) splits $\Lambda = -1 \star (\mu \cdot \deg)$ into a small Möbius divisor and the complement of a large one: two columns. The paper’s separate Vaughan Type I/II decomposition occurs in Appendix A. Standard Type I/II regrouping likewise has a short-variable part and a large–large bilinear part [\[9, §VI\]](#); it does not supply a canonical third large-divisor column. A small/middle/large spectral sequence must combine an outer divisor-degree filtration with this inner cubical refinement.

Research Target 4.25 (Marked cubical Type I/II comparison). *For every j, U , upgrade the K_0 cancellation*

$$[\mathcal{P} \star \mathcal{Q}_U^{*j}] = [\mathcal{L} \star \mathcal{Z}^{*(j-1)} \star \mathcal{M}_{\leq U}^{*j}]$$

to a filtered comparison, or to an explicit common filtered replacement, compatible up to coherent homotopy with every cubical face map. Its first extension map must be the connection $\beta_a \epsilon_b$ of [theorem 4.18](#). After degree-box localization, identify the northeast range with a Type II object and prove the required weight–Betti or alternating-trace estimates.

5. THE UNIVERSAL TWO-FORM CORRELATION

Let $\mathbf{H}_d \simeq \mathbb{A}_{\mathbb{F}_q}^d$ parametrize polynomials h of degree $< d$. On

$$X_d := \mathbf{M}_d \times \mathbf{H}_d$$

consider the two affine-linear maps

$$L_1(f, h) = f, \quad L_2(f, h) = f + h$$

to $\mathbf{M}_d$, and let $p : X_d \rightarrow \mathbf{H}_d$ be the second projection.

Definition 5.1 (Relative twin-correlation complex). Define

$$(5.1) \quad \mathcal{T}_d := Rp_! \left(L_1^* \mathcal{P}_d \overset{L}{\otimes} L_2^* \mathcal{P}_d \right) \in D_c^b(\mathbf{H}_d, \overline{\mathbb{Q}}_\ell).$$

Theorem 5.2 (Relative trace identity). *For every $h \in \mathbf{H}_d(\mathbb{F}_q)$,*

$$(5.2) \quad \mathrm{Tr}^{\mathrm{alt}}(\mathrm{Frob}_h, (\mathcal{T}_d)_{\bar{h}}) = \sum_{f \in \mathbf{M}_d(\mathbb{F}_q)} \Lambda(f) \Lambda(f + h).$$

Proof. Proper base change for $p_!$ identifies the stalk at h with compactly supported cohomology over the f -fiber. The Grothendieck–Lefschetz trace formula turns its alternating Frobenius trace into the sum over $f \in \mathbf{M}_d(\mathbb{F}_q)$ of the pointwise trace of the tensor product. Traces multiply under derived tensor product, and [theorem 4.1](#) identifies the two factors with $\Lambda(f)$ and $\Lambda(f + h)$. $\square$

The same construction is not restricted to twins. If $g : X \rightarrow B$ is a separated morphism of finite type and $L_1, L_2 : X \rightarrow \mathbf{M}_d$ are any two morphisms, set

$$\mathcal{T}(X/B; L_1, L_2) := Rg_! \left(L_1^* \mathcal{P}_d \overset{L}{\otimes} L_2^* \mathcal{P}_d \right).$$

Its trace at $b \in B(\mathbb{F}_q)$ is the sum of $\Lambda(L_1(x))\Lambda(L_2(x))$ over $x \in X_b(\mathbb{F}_q)$. Thus the prime complex separates the formal prime-detection step from the geometry of the particular family of linear forms.

5.1. The relative Goldbach object. Let $n \geq 2$, put $Y_n = \mathbf{M}_{n-1} \times \mathbf{M}_n$, and let $r : Y_n \rightarrow \mathbf{M}_n$ be the second projection. The maps

$$K_1(g, F) = g \in \mathbf{M}_{n-1}, \quad K_2(g, F) = F - g \in \mathbf{M}_n$$

are well defined: because $\deg g = n - 1$, the polynomial $F - g$ remains monic of degree n . Define

$$(5.3) \quad \mathcal{G}_n := Rr_! \left(K_1^* \mathcal{P}_{n-1} \overset{L}{\otimes} K_2^* \mathcal{P}_n \right) \in D_c^b(\mathbf{M}_n, \overline{\mathbb{Q}}_\ell).$$

The same base-change argument as in [theorem 5.2](#) gives the exact identity

$$(5.4) \quad \mathrm{Tr}^{\mathrm{alt}}(\mathrm{Frob}_F, (\mathcal{G}_n)_{\bar{F}}) = \sum_{g \in \mathbf{M}_{n-1}(\mathbb{F}_q)} \Lambda(g) \Lambda(F - g).$$

This constructs the Goldbach object but does not estimate it. The difference between (5.3) and the shift family (5.1) is concentrated at the infinite place; it is made precise in [section 9](#).

6. INCIDENCE EXPANSION AND THE AFFINE-BUNDLE RANGE

Expanding (5.1) gives a concrete incidence model. For $1 \leq a, b \leq d$, let

$$Z_{a,b,d} := \{(D, E, f, h) \in \mathbf{U}_a \times \mathbf{U}_b \times \mathbf{M}_d \times \mathbf{H}_d : D \mid f, E \mid f + h\},$$

and write $\pi_{a,b,d} : Z_{a,b,d} \rightarrow \mathbf{H}_d$ for the projection. The pullback of $\mathcal{M}_a \boxtimes \mathcal{M}_b$ to $Z_{a,b,d}$ will be denoted by the same symbol. Base change and the projection formula give

$$(6.1) \quad \mathcal{T}_d \simeq \bigoplus_{a,b=1}^d (R\pi_{a,b,d,!}(\mathcal{M}_a \boxtimes \mathcal{M}_b)[2])^{\oplus ab}.$$

The even shift $[2]$ records the two shifts used to pass from $\mathcal{A}_d$ to $\mathcal{P}_d$.

Stratify $\mathbf{U}_a \times \mathbf{U}_b$ by

$$c = \deg \gcd(D, E), \quad r = a + b - c = \deg \operatorname{lcm}(D, E).$$

The condition $\deg \gcd(D, E) \geq c$ is closed, for example by the vanishing of the appropriate subresultants, so this is a finite algebraic stratification and gives the usual compact-support spectral sequence.

Proposition 6.1 (Affine-bundle range). *Fix a geometric triple (D, E, h) in the degree- c gcd stratum and put $G = \gcd(D, E)$ and $r = \deg \operatorname{lcm}(D, E)$.*

(1) *The congruences*

$$f \equiv 0 \pmod{D}, \quad f \equiv -h \pmod{E}$$

are compatible if and only if $G \mid h$.

(2) *If $G \mid h$ and $r \leq d$, the monic degree- d solutions form an affine space of dimension $d - r$.*

(3) *If $r > d$, there is at most one monic degree- d solution.*

Over the constant-rank stratum with $G \mid h$ and $r \leq d$, these affine spaces form an affine bundle of rank $d - r$.

Proof. The first assertion is the Chinese remainder theorem. If it holds, the two congruences specify a unique residue class R modulo the monic polynomial $L = \operatorname{lcm}(D, E)$. When $r \leq d$, every monic degree- d solution is uniquely

$$f = R + LQ,$$

where Q is monic of degree $d - r$. Such Q form $\mathbb{A}^{d-r}$. Equivalently, the congruence equations are a family of linear equations of constant rank r , so their solution spaces form a torsor under a rank- $(d - r)$ vector bundle. If $r > d$, the difference of two solutions would have degree $< r$ and be divisible by L , hence would vanish. $\square$

Remark 6.2 (The first algebraic boundary). The inequality

$$\deg \operatorname{lcm}(D, E) \leq d$$

is exactly the range in which the incidence geometry has the expected affine dimension. Beyond it, existence is an additional condition on (D, E, h) and the affine-bundle calculation stops. This provides a geometric boundary before any analytic estimate is invoked.

For each (a, b) , let $i_{a,b} : Z_{a,b,d}^{\text{st}} \hookrightarrow Z_{a,b,d}$ be the closed locus $r \leq d$ and let $j_{a,b} : Z_{a,b,d}^{\text{un}} \hookrightarrow Z_{a,b,d}$ be its open complement. Put

$$\mathcal{K}_{a,b} := (\mathcal{M}_a \boxtimes \mathcal{M}_b)|_{Z_{a,b,d}}[2]$$

and define

$$\mathcal{T}_d^{\text{st}} := \bigoplus_{a,b=1}^d \left(R\pi_{a,b,d,!} i_{a,b,*} i_{a,b}^* \mathcal{K}_{a,b} \right)^{\oplus ab},$$

$$\mathcal{T}_d^{\text{un}} := \bigoplus_{a,b=1}^d \left(R\pi_{a,b,d,!} j_{a,b,!} j_{a,b}^* \mathcal{K}_{a,b} \right)^{\oplus ab}.$$

The open-closed triangle gives a functorial exact triangle

$$(6.2) \quad \mathcal{T}_d^{\text{un}} \longrightarrow \mathcal{T}_d \longrightarrow \mathcal{T}_d^{\text{st}} \longrightarrow \mathcal{T}_d^{\text{un}}[1].$$

Corollary 6.3 (Stable trace). *For $h \in \mathbf{H}_d(\mathbb{F}_q)$,*

$$(6.3) \quad \text{Tr}^{\text{alt}}(\text{Frob}_h, (\mathcal{T}_d^{\text{st}})_{\bar{h}}) = q^d \sum_{\substack{D,E \text{ monic} \\ (D,E)|h \\ \deg \text{lcm}(D,E) \leq d}} \frac{\mu(D)\mu(E) \deg D \deg E}{|\text{lcm}(D,E)|}.$$

Proof. Only squarefree D, E contribute. For each compatible pair in the stable range, [proposition 6.1](#) gives exactly $q^{d-\deg \text{lcm}(D,E)}$ rational points in the f -fiber. The two Möbius objects contribute $\mu(D)\mu(E)$, and the multiplicities in [\(6.1\)](#) contribute $\deg D \deg E$. $\square$

7. THE EULER PRODUCT AND THE SINGULAR-SERIES SUMMAND

Introduce the three-variable density series

$$(7.1) \quad \Phi_h(u, v, w) := \sum_{\substack{D,E \text{ monic} \\ (D,E)|h}} \frac{\mu(D)\mu(E) u^{\deg D} v^{\deg E} w^{\deg \text{lcm}(D,E)}}{|\text{lcm}(D,E)|}.$$

It is initially interpreted in a small polydisc. Since μ restricts the sum to squarefree polynomials, its Euler factors are

$$(7.2) \quad \Phi_{h,P}(u, v, w) = \begin{cases} 1 - \frac{(uw)^{\deg P}}{|P|} - \frac{(vw)^{\deg P}}{|P|}, & P \nmid h, \\ 1 - \frac{(uw)^{\deg P}}{|P|} - \frac{(vw)^{\deg P}}{|P|} + \frac{(uvw)^{\deg P}}{|P|}, & P \mid h. \end{cases}$$

For monic irreducible P , put

$$U_P = \frac{(uw)^{\deg P}}{|P|}, \quad V_P = \frac{(vw)^{\deg P}}{|P|}.$$

Using

$$\prod_P (1 - U_P) = 1 - uw, \quad \prod_P (1 - V_P) = 1 - vw,$$

we obtain

$$(7.3) \quad \Phi_h(u, v, w) = (1 - uw)(1 - vw)G_h(u, v, w),$$

where G_h is analytic in a neighborhood of $(1, 1, 1)$. Indeed, away from the finitely many primes dividing h , the quotient local factor differs from 1 by

$$-\frac{U_P V_P}{(1 - U_P)(1 - V_P)},$$

whose prime sum converges when $|uvw^2| < q$ in a sufficiently small polydisc around $(1, 1, 1)$.

Let $D_u = u \frac{\partial}{\partial u}$ and $D_v = v \frac{\partial}{\partial v}$. Write

$$(7.4) \quad A_h(w) := D_u D_v \Phi_h(u, v, w)|_{u=v=1} = \sum_{r \geq 0} A_r(h) w^r.$$

Then $A_r(h)$ is exactly the inner density in [\(6.3\)](#) with $\deg \text{lcm}(D, E) = r$.

Proposition 7.1 (Singular-series extraction). *For every fixed nonzero h ,*

$$A_h(1) = G_h(1, 1, 1) = \mathfrak{S}_q(h),$$

where

$$\mathfrak{S}_q(h) = \prod_{P|h} (1 - |P|^{-1})^{-1} \prod_{P \nmid h} (1 - (|P| - 1)^{-2}).$$

Moreover, there exists $R = R(q) > 1$ such that, for fixed h ,

$$\sum_{r>d} A_r(h) = O_{q,h}(R^{-d}).$$

Proof. Differentiate (7.3). At $u = v = w = 1$, each factor $1 - uw$ and $1 - vw$ vanishes, so the only surviving term is obtained by differentiating each of them once. Thus $A_h(1) = G_h(1, 1, 1)$.

If $P \nmid h$, the quotient of the first local factor in (7.2) by $(1 - |P|^{-1})^2$ is

$$\frac{1 - 2|P|^{-1}}{(1 - |P|^{-1})^2} = 1 - (|P| - 1)^{-2}.$$

If $P | h$, the corresponding quotient is $(1 - |P|^{-1})^{-1}$. This gives the displayed Euler product. Analyticity in a polydisc of radius greater than 1 and Cauchy's estimate give exponential decay of the w -coefficients. $\square$

7.1. A one-variable bar zeta function. The same constant admits a first-derived-value description closer to the L -function language. The calculation below is a fixed- h restatement of [23, Proposition 6.2], included to make the bar interface explicit. Let $\varphi(D) = |(\mathbb{F}_q[T]/D)^\times|$ and, initially for $|u| < 1$, set

$$(7.5) \quad Z_h^{\text{bar}}(u) := \sum_{\substack{D \text{ monic} \\ (D, h)=1}} \frac{\mu(D) u^{\deg D}}{\varphi(D)} = \prod_{P \nmid h} \left(1 - \frac{u^{\deg P}}{|P| - 1} \right).$$

The coefficient $\mu(D)$ is the trace of the interval object $\mathcal{M}_{\deg D}$; thus (7.5) is the numerical realization of a virtual bar object, not an additional prime-detection identity.

Proposition 7.2 (Sawin–Shusterman bar-zeta derivative). *For fixed nonzero h , the function in (7.5) has a factorization, giving its analytic continuation to a neighborhood of $u = 1$,*

$$Z_h^{\text{bar}}(u) = (1 - u)H_h(u),$$

where H_h is analytic near $u = 1$ and

$$H_h(1) = \mathfrak{S}_q(h), \quad -u \frac{d}{du} Z_h^{\text{bar}}(u) \Big|_{u=1} = \mathfrak{S}_q(h).$$

Proof. Divide the Euler product in (7.5) by

$$\prod_P \left(1 - \frac{u^{\deg P}}{|P|} \right) = 1 - u.$$

For $P \nmid h$, the quotient local factor at $u = 1$ is

$$\frac{1 - (|P| - 1)^{-1}}{1 - |P|^{-1}} = 1 - (|P| - 1)^{-2},$$

while for $P | h$ it is $(1 - |P|^{-1})^{-1}$. Away from the finitely many primes dividing h , the quotient differs from 1 by $O(|P|^{-2})$, so the product is analytic near 1. Its value is exactly the Euler product defining $\mathfrak{S}_q(h)$ in proposition 7.1. Differentiating $(1 - u)H_h(u)$ proves the second identity. $\square$

Remark 7.3. The factor $1/\varphi(D)$ suggests a stack-theoretic realization through the classifying stack of $(\mathbb{F}_q[T]/D)^\times$, whose groupoid cardinality is $1/\varphi(D)$. Constructing the required completed motivic or ℓ -adic object, and proving that its degree derivative agrees with the filtered prime complex rather than only with its trace, remains a separate completed Euler-product problem. The finite marked connection of [theorem 4.18](#) does not by itself construct this stacky completion.

Corollary 7.4 (Localization of the parity remainder). *For fixed nonzero h ,*

$$(7.6) \quad \sum_{f \in \mathbf{M}_d(\mathbb{F}_q)} \Lambda(f) \Lambda(f+h) - \mathfrak{S}_q(h) q^d = \mathrm{Tr}^{\mathrm{alt}}(\mathrm{Frob}_h, (\mathcal{T}_d^{\mathrm{un}})_{\bar{h}}) - q^d \sum_{r>d} A_r(h).$$

The second term on the right is $O_{q,h}(q^d R^{-d})$.

Proof. Combine [theorem 5.2](#), [corollary 6.3](#), and [proposition 7.1](#) and use the additivity of traces in the open–closed triangle. $\square$

Remark 7.5. [Equation \(7.6\)](#) is an exact localization statement. It does not prove that the unstable trace is small. It says that, after the affine-bundle range and its Euler product have been removed, the complete twin-prime problem is concentrated in the explicitly defined unstable complex, together with the separate analytic tail $q^d \sum_{r>d} A_r(h)$.

8. THE SPECTRAL SEQUENCE AND THE MISSING WEIGHT THEOREM

For fixed a, b , the closed filtration by $\deg \gcd(D, E) \geq c$ yields a compact-support spectral sequence whose E_1 -terms are the derived pushforwards from the exact gcd strata. On a stratum with $r = a + b - c \leq d$, [proposition 6.1](#) gives the affine-bundle identity

$$(8.1) \quad R\rho_! \overline{\mathbb{Q}}_\ell \simeq \overline{\mathbb{Q}}_\ell(-(d-r))[-2(d-r)].$$

Consequently the stable E_1 -terms reduce to cohomology of configuration-type bases with the two sign local systems and an explicit Tate twist. The terms with $r > d$ are precisely the non-affine terms.

This has the same general architecture as the bar-complex spectral sequences in [\[6\]](#): a downward-closed, expected-dimension range is computable, while the complement must be controlled cohomologically. The geometry here is simpler, but the arithmetic coefficient $\mathcal{P}_d$ and the linear growth of the truncation range create a different weight problem.

Research Target 8.1 (Residual weight–Betti bound). *Choose a suitable nondegenerate shift stratum $\mathbf{H}_d^c \subset \mathbf{H}_d$ and set*

$$\mathcal{R}_d := \mathcal{T}_d^{\mathrm{un}}|_{\mathbf{H}_d^c}.$$

Prove constants $\eta > 0$ and $C \geq 1$ such that

(1) *every Frobenius eigenvalue on $H^i((\mathcal{R}_d)_{\bar{h}})$ has weight at most $2(1-\eta)d$, and*

(2) $\sum_i \dim H^i((\mathcal{R}_d)_{\bar{h}}) \leq C^d$

uniformly in the shift parameter.

If $q^\eta > C$, the target would imply, with $\delta = \eta - \log_q C > 0$,

$$\mathrm{Tr}^{\mathrm{alt}}(\mathrm{Frob}_h, (\mathcal{R}_d)_{\bar{h}}) = O\left(q^{(1-\delta)d}\right).$$

This is the precise point at which a homological construction would produce arithmetic cancellation. Combining it with the already separate exponential estimate for $q^d \sum_{r>d} A_r(h)$ in [corollary 7.4](#) would bound the full error term.

Sawin and Shusterman already prove the required arithmetic trace bound by analytic and geometric estimates under an explicit large-field hypothesis [\[23, Theorem 6.3\]](#). Sawin’s later estimate for squarefree progressions [\[21\]](#) improves the small-divisor input. Those results certify that the desired trace cancellation is true in a range of q , but they do not by themselves give the cohomological weight–Betti statement in [research Target 8.1](#). Establishing such

a statement, or deriving the known trace bound from a substantially shorter comparison theorem for $\mathcal{R}_d$, is the first genuinely algebraic objective.

9. THE GOLDBACH BOUNDARY AND THE INFINITE PLACE

The degree pattern in (5.4), with summands of degrees $n - 1$ and n , is the polynomial Goldbach problem studied by Bender–Pollack [3]; they obtain an asymptotic when q is sufficiently large relative to n . Thus (5.3) is a cohomological packaging of an existing problem, not a new formulation of Goldbach. The possible new target below concerns a power-saving regime with fixed sufficiently large q and $n \rightarrow \infty$.

The exact object (5.3) lets us state that target column by column. Put $d = n - 1$. Expanding the first von Mangoldt factor gives

$$(9.1) \quad \sum_{g \in \mathbf{M}_d(\mathbb{F}_q)} \Lambda(g) \Lambda(F - g) = - \sum_{k=1}^d k \sum_{D \in \mathbf{M}_k(\mathbb{F}_q)} \mu(D) \sum_{N \in \mathbf{M}_{d-k}(\mathbb{F}_q)} \Lambda(F - DN).$$

This is the alternating trace of the divisor-degree columns of $\mathcal{G}_n$; by itself it is still the usual Möbius decomposition.

For D squarefree and coprime to F , the small-divisor input suggested by (9.1) is a uniform estimate of the form

$$(9.2) \quad \sum_{N \in \mathbf{M}_{d-k}(\mathbb{F}_q)} \Lambda(F - DN) = \frac{q^d}{\varphi(D)} + O\left(q^{(d-k)(1-\delta)}\right).$$

This is not literally the squarefree-progression estimate in [21]. Indeed, $H = F - DN$ is monic of degree n , satisfies $H \equiv F \pmod{D}$, and has prescribed coefficient

$$[T^{n-1}]H = [T^{n-1}]F - 1.$$

Thus, in addition to the finite modulus D , one has fixed one extra coefficient at the infinite place. In Hayes/ray-class language the natural modulus is $D\infty^2$. More explicitly, for the reciprocal polynomial $A^*(U) = U^{\deg A} A(U^{-1})$ one has

$$H^*(U) = F^*(U) - UD^*(U)N^*(U),$$

so the congruence modulo U^2 records exactly this extra coefficient.

For k in the large-divisor range, the corresponding parity-breaking input is

$$(9.3) \quad \sum_{D \in \mathbf{M}_k(\mathbb{F}_q)} \mu(D) \Lambda(F - DN) \ll q^{k(1-\eta)}$$

uniformly in monic N of degree $d - k$. Here

$$\deg F = k + \deg N + 1.$$

In the notation of [23, Corollary 6.1], the large-divisor estimate requires the fixed additive term to have degree strictly less than the sum of the two variable degrees. Matching the Goldbach column would require $\deg F < k + \deg N$, whereas here $\deg F = k + \deg N + 1$. The resulting degree obstruction manifests after reciprocity as the ∞^2 condition above. Other hypotheses must also be audited: in particular, one must separate $(F, N) > 1$ and handle the sign and leading-coefficient normalization when replacing N by $-N$.

Research Target 9.1 (Ray-class Goldbach columns). *Extend the squarefree-progression and parity-breaking estimates of [21, 23] to the bounded additional local condition at ∞^2 , proving (9.2) for $k \leq \theta d$ and (9.3) for $k > \theta d$, for some $1/2 < \theta < 1$ and $\delta, \eta > 0$ in an explicit field range. Categorically, construct the corresponding ray-class local system and prove the estimates as uniform Frobenius-trace bounds for the relevant columns of $\mathcal{G}_n$, including the coprimality and normalization strata just noted.*

Once [research Target 9.1](#) is proved with summable uniform constants, Möbius summation in (9.1) and the same bar-zeta extraction as in [proposition 7.2](#) yield the usual Goldbach singular series times q^{n-1} , with a power-saving error. This implication is formal; the target itself is the new arithmetic input. Its advantage is that it identifies a bounded-conductor local modification as the first specific obstruction. It does not assert that the existing estimates survive that modification without further geometric work.

10. THE FOURTH RAY-CLASS STEP: FOURIER AND VANISHING CYCLES

This section carries out the formal and microlocal parts of the first estimate in [research Target 9.1](#). It proves the complete nonzero-frequency estimate in every characteristic and obtains a fixed- q , all-degree range crossing $1/2$. It does six things:

- (1) it defines the filtered ray object whose nonzero Fourier stalk is exactly the required additive twist;
- (2) it proves, in arbitrary characteristic, which of Sawin's characteristic-cycle components can meet a nonzero infinity frequency; and
- (3) it gives one simultaneous normal form on every ordered boundary stratum and proves that all relative ray-parameter vanishing cycles there vanish;
- (4) it proves fixed-frequency perversity and two-degree concentration;
- (5) it computes the top polar multiplicity in arbitrary characteristic and the generic rank by a true local-intersection calculation; and
- (6) it converts the two exact multiplicities into a fixed- q bounded-additive-sieve estimate beyond the $1/2$ threshold.

Thus, for squarefree D and conductor two at infinity, neither local boundary classification nor the special-stalk polar bound remains open. The nonreduced multiplicity of a wild fixed cycle is retained by the true local term; replacing it by a naive reduced-point count would give the wrong generic rank.

10.1. The ray coordinate and its Fourier stalk. Fix a nontrivial additive character $\psi_q = \psi \circ \text{Tr}_{\mathbb{F}_q/\mathbb{F}_p}$. For a squarefree monic $D \in \mathbb{F}_q[T]$ of degree k , put

$$A_D = \mathbb{F}_q[T]/(D), \quad \mathbb{G}_D = \text{Res}_{A_D/\mathbb{F}_q} \mathbb{G}_m, \quad \mathbb{R}_D = \mathbb{G}_D \times \mathbb{G}_a.$$

Thus $\mathbb{G}_D(\mathbb{F}_q) = A_D^\times$. Write

$$f = T^n + c_1(f)T^{n-1} + \dots$$

for $f \in \mathbf{M}_n$. On $X_{n,D} = \{f \in \mathbf{M}_n : (f, D) = 1\}$ define

$$(10.1) \quad \rho_{n,D}^{(1)} : X_{n,D} \longrightarrow \mathbb{R}_D, \quad f \longmapsto (f \bmod D, c_1(f)).$$

The second coordinate is genuinely a ray coordinate, rather than an ad hoc additive statistic. For monic polynomials,

$$c_1(fg) = c_1(f) + c_1(g),$$

and, with $f^*(U) = U^{\deg f} f(U^{-1})$,

$$f^*(U) \equiv 1 + c_1(f)U \pmod{U^2}.$$

Consequently (10.1) is the monic, fixed-degree chart of the ray group for $D\infty^2$. The group law on the displayed chart is $(a, t)(a', t') = (aa', t + t')$.

Let $\mathcal{K}_n^{\text{bar}}$ carry the divisor-degree filtration G from [theorems 4.5](#) and [4.11](#), and define the filtered pushforward

$$\mathcal{E}_{n,D}^{\text{fil}} := R\rho_{n,D,!}^{(1)}(\mathcal{K}_n^{\text{bar}}|_{X_{n,D}}, G) \quad \text{on } \mathbb{G}_D \times \mathbb{A}_t^1.$$

Use the unnormalized arithmetic Fourier transform

$$(10.2) \quad \Phi_{n,D}^{\text{fil}} := Rp_{13,!} \left(p_{12}^* \mathcal{E}_{n,D}^{\text{fil}} \otimes \mathcal{L}_{\psi_q(t\alpha)} \right) \quad \text{on } \mathbb{G}_D \times \mathbb{A}_\alpha^1.$$

Here the projections have source $\mathbb{G}_D \times \mathbb{A}_t^1 \times \mathbb{A}_\alpha^1$. No perverse normalizing shift has been inserted; inserting [1] would change the sign of the alternating trace below.

Proposition 10.1 (Fourier trace and ray inversion). *For $a \in A_D^\times$ and $\alpha \in \mathbb{F}_q$, set*

$$S_\alpha(n; D, a) := \sum_{\substack{f \in \mathbf{M}_n(\mathbb{F}_q) \\ f \equiv a \pmod{D}}} \Lambda(f) \psi_q(\alpha c_1(f)).$$

Then

$$\mathrm{Tr}^{\mathrm{alt}}\left(\mathrm{Frob}_{(a,\alpha)}, (\Phi_{n,D}^{\mathrm{fil}})_{(a,\alpha)}\right) = S_\alpha(n; D, a).$$

If

$$A_{a,t} := \sum_{\substack{f \in \mathbf{M}_n(\mathbb{F}_q) \\ f \equiv a \pmod{D}, c_1(f)=t}} \Lambda(f),$$

then

$$(10.3) \quad S_\alpha(n; D, a) = \sum_{t \in \mathbb{F}_q} A_{a,t} \psi_q(\alpha t), \quad A_{a,t} = \frac{1}{q} \sum_{\alpha \in \mathbb{F}_q} S_\alpha(n; D, a) \psi_q(-\alpha t).$$

The construction and both identities are compatible with the divisor-degree filtration term by term.

Proof. Proper base change identifies the stalk of (10.2) with compactly supported cohomology of the fiber $f \equiv a \pmod{D}$, tensored with $c_1^* \mathcal{L}_{\psi_q(\alpha \cdot)}$. The trace of $\mathcal{K}_n^{\mathrm{bar}}$ is Λ , so Grothendieck–Lefschetz gives the first identity. The other two are finite Fourier inversion on the additive group of $\mathbb{F}_q$. All functors used in (10.2) preserve finite filtered quasi-isomorphisms, which gives the last assertion. $\square$

Definition 10.2 (The first bounded-additive-sieve estimate). For $0 < \theta < 1$ and $\delta > 0$, write $\mathrm{BAS}_1(\theta, \delta)$ for the assertion that

$$(10.4) \quad |S_\alpha(n; D, a)| \leq C_{q,\theta} (q^{n-k})^{1-\delta}$$

uniformly for squarefree D of degree $k \leq \theta n$, $a \in A_D^\times$, and $\alpha \in \mathbb{F}_q^\times$.

Together with the $\alpha = 0$ squarefree-progression estimate, [proposition 10.1](#) turns (10.4) into a fixed- c_1 ray-class estimate. The difference between $n - k$ and the actual fixed-coefficient fiber dimension $n - k - 1$ is one fixed power of q , and is absorbed in the constant when q is fixed. In the Goldbach substitution $n = d + 1$, $a = F \pmod{D}$, and $t = c_1(F) - 1$, the map

$$N \mapsto H = F - DN$$

is a bijection from $\mathbf{M}_{d-k}$ to the corresponding ray fiber. Its main term is $q^{n-1}/\varphi(D) = q^d/\varphi(D)$, exactly as in (9.2).

10.2. What Riemann hypothesis alone gives. The ordinary explicit formula already gives a useful baseline, but it also displays the $1/2$ barrier exactly.

Proposition 10.3 (Pointwise and mean-square baselines). *For $\alpha \neq 0$,*

$$(10.5) \quad |S_\alpha(n; D, a)| \leq (k+2)q^{n/2} + 1,$$

$$(10.6) \quad \sum_{a \in A_D^\times} |S_\alpha(n; D, a)|^2 \leq ((k+2)q^{n/2} + 1)^2.$$

Consequently, for $\theta < 1/2$, (10.4) follows for every

$$0 < \delta < \frac{1-2\theta}{2(1-\theta)}.$$

For every $\epsilon > 0$ and fixed nonzero α , all but at most $\epsilon\varphi(D)$ residue classes a satisfy

$$(10.7) \quad |S_\alpha(n; D, a)| \leq \epsilon^{-1/2} \frac{(k+2)q^{n/2} + 1}{\sqrt{\varphi(D)}}.$$

Proof. Expand the condition $f \equiv a \pmod{D}$ in multiplicative characters χ of $A_D^\times$. The product

$$f \mapsto \chi(f)\psi_q(\alpha c_1(f))$$

is a rank-one ray character of conductor dividing $D\infty^2$. Since $\alpha \neq 0$, it is geometrically nontrivial at infinity. The explicit formula and the Riemann hypothesis over finite fields bound its degree- n von Mangoldt sum by $((k+2)q^{n/2} + 1)$; the harmless $k+2$ is a uniform upper bound for the degree of the relevant L -polynomial [8]. Character orthogonality gives (10.5). Plancherel on $A_D^\times$ gives

$$\sum_a |S_\alpha(n; D, a)|^2 = \frac{1}{\varphi(D)} \sum_\chi \left| \sum_{\substack{f \in \mathbf{M}_n \\ (f, D)=1}} \Lambda(f) \chi(f) \psi_q(\alpha c_1(f)) \right|^2,$$

which proves (10.6). Comparing $q^{n/2}$ with $q^{(n-k)(1-\delta)}$ gives the stated range when $k \leq \theta n$; the polynomial factor in n is absorbed by a slight decrease of δ . Finally, (10.7) is Chebyshev's inequality. $\square$

Remark 10.4. The almost-all estimate is square-root size in the ray fiber and is a classical Fourier- L -function consequence. Hybrid variances for primes in short residue classes are studied more precisely by Baier–Bhandari [2]. Neither (10.6) nor those averaged results control the particular Goldbach residue $a = F \pmod{D}$. A worst-class estimate beyond $1/2$ requires new relative information.

10.3. A characteristic-free interior cutoff. The relevant characteristic-cycle components are not always ordinary conormals to multiplicity strata. This distinction is essential when the characteristic divides a root multiplicity. We therefore use Sawin's cycles themselves [20, Definition 2.8 and Theorem 2.10]; see also [21, Lemma 6.3].

Let P_r denote the vector space of polynomials of degree $< r$. For a weight function w with

$$\sum_{j \geq 1} jw(j) = n, \quad s = s(w) := \sum_{j \geq 1} w(j),$$

a point of $\prod_j C^{(w(j))}$ is represented by monic polynomials f_j of degree $w(j)$. Put

$$h_w := \prod_{j \geq 1} f_j^{j-1}, \quad \deg h_w = n - s.$$

The vector bundle defining Sawin's cycle B_w has fiber

$$(10.8) \quad W_w = \ker((\cdot h_w)^t : P_n^\vee \longrightarrow P_s^\vee).$$

Formula (10.8), not the possibly inseparable differential of $(f_j) \mapsto \prod_j f_j^j$, is the correct object in arbitrary characteristic.

Theorem 10.5 (Nonzero infinity-frequency cutoff). *Assume $(h_w, D) = 1$. Define*

$$L_{h_w} : P_s \longrightarrow P_k \oplus \mathbb{F}_q, \quad q_0 \longmapsto (h_w q_0 \pmod{D}, [T^{n-1}] h_w q_0).$$

Then:

- (1) *if $s \geq k+1$, the map L_{h_w} is surjective, so the image of a nonzero base covector under the transpose of $d(r_D, c_1)$ does not meet W_w ;*
- (2) *if $s \leq k$, then for every $\alpha \neq 0$ there is $\lambda \in P_k^\vee$ such that*

$$L_{h_w}^t(\lambda, \alpha) = 0.$$

Consequently the interior canonical relation of (10.2), over $\alpha \neq 0$, meets B_w if and only if $s(w) \leq k$. This statement is valid in every characteristic.

Proof. Since h_w is monic of degree $n - s$,

$$[T^{n-1}]h_w q_0 = [T^{s-1}]q_0.$$

Suppose first that $s \geq k + 1$. Given $(u, c) \in P_k \oplus \mathbb{F}_q$, there is a unique $q_1 \in P_k$ with $h_w q_1 \equiv u \pmod{D}$, because h_w is a unit modulo D . Write $q_0 = q_1 + tD$ with $t \in P_{s-k}$. The first coordinate is unchanged, while

$$[T^{s-1}](tD) = [T^{s-k-1}]t.$$

Choosing the last coefficient of t gives any prescribed second coordinate. Thus L_{h_w} is surjective.

Now suppose $s \leq k$. The map $q_0 \mapsto h_w q_0 \pmod{D}$ is injective: if its value vanishes, then $D \mid q_0$, while $\deg q_0 < s \leq k = \deg D$. For a prescribed $\alpha \neq 0$, define a linear functional on its image by

$$\lambda(h_w q_0 \pmod{D}) = -\alpha [T^{s-1}]q_0$$

and extend it to P_k . Then $L_{h_w}^t(\lambda, \alpha) = 0$.

A base covector belongs to (10.8) exactly when it annihilates $h_w P_s$, which is exactly the displayed transpose condition. This proves the canonical-relation assertion. $\square$

Remark 10.6 (Tame coordinates and the wild correction). If the characteristic is larger than n , write $f = \prod_{u=1}^s (T - x_u)^{m_u}$ and split $D = \prod_{j=1}^k (T - d_j)$. The theorem becomes the elementary system

$$\alpha + \sum_{j=1}^k \frac{b_j}{d_j - x_u} = 0 \quad (1 \leq u \leq s).$$

The Cauchy matrix has rank $\min(s, k)$, and after multiplication by $D(X)$ the left side has a degree- k numerator with leading coefficient α . This gives the same cutoff.

When $p \mid m_u$, that tangent calculation is not valid: for example $c_1((T - x)^p) = 0$. There is no contradiction with theorem 10.5, because in characteristic p the cycle B_w need not be the conormal bundle of its set-theoretic support. The ideal-theoretic bundle (10.8) is precisely what repairs the wild case.

Corollary 10.7 (Filtered form). *For every graded piece of the Euler-tail filtration on $\mathcal{K}_n^{\text{bar}}$, all affine-interior contributions to $\Phi_{n,D}^{\text{fil}}|_{\alpha \neq 0}$ come from weight functions with $s(w) \leq k$. Any further nonzero-frequency contribution must come from the boundary of a compactification of $\rho_{n,D}^{(1)}$.*

Proof. The factorization sheaves in the graded pieces are direct summands of the symmetric-power sheaves to which Sawin's characteristic-cycle formula applies. Direct summands can only shrink singular support. Apply theorem 10.5 term by term. The final qualification is necessary because an affine phase without an interior critical point can still acquire vanishing cycles at infinity. $\square$

10.4. The coefficient equation at infinity. We next calculate the boundary term which is new relative to a squarefree finite modulus. Suppose $r \geq 2$ ordered roots tend to infinity and write $x_i = 1/y_i$. After the finite roots have been kept in a fixed affine chart, the equation $c_1(f) = b$ has the form

$$(10.9) \quad F_B(y) := B \prod_{i=1}^r y_i + \sum_{i=1}^r \prod_{\ell \neq i} y_\ell = 0,$$

where $B = b + \sum x_{\text{finite}}$ is regular in the chart. One root at infinity cannot occur in the closure over finite b , because for $r = 1$ the left side is $1 + By_1$ at $y_1 = 0$.

Lemma 10.8 (Pure-infinity stationary phase). *Near $y_1 = \cdots = y_r = 0$, the family $F_B = 0$ is isomorphic over the B -line to a product with that line. More precisely, on $1 + By_1 \neq 0$ put*

$$z_1 = \frac{y_1}{1 + By_1}, \quad z_i = y_i \quad (i > 1).$$

Then $F_B = 0$ becomes

$$(10.10) \quad \prod_{i=2}^r z_i + z_1 \sum_{i=2}^r \prod_{\substack{2 \leq \ell \leq r \\ \ell \neq i}} z_\ell = 0,$$

independently of B . Hence the vanishing cycles in the B direction of the extension by zero from $\prod_i y_i \neq 0$ vanish.

Proof. The inverse change is $y_1 = z_1/(1 - Bz_1)$ and is defined near the boundary. Substitution gives

$$F_B = \frac{1}{1 - Bz_1} \left(\prod_{i=2}^r z_i + z_1 \sum_{i=2}^r \prod_{\substack{2 \leq \ell \leq r \\ \ell \neq i}} z_\ell \right).$$

The prefactor is a unit. The same coordinate change carries $\prod_i y_i \neq 0$ to $\prod_i z_i \neq 0$. Thus both the hypersurface and its open complement are pulled back from the B -independent fiber, so extension by zero is universally locally acyclic over the B -line. $\square$

The singularity in (10.10) can be non-smooth for $r \geq 3$, but it is constant in the coefficient direction; singularity by itself is not the obstruction. The calculation also shows why the Artin–Schreier factor is harmless on the coefficient graph: it is the pullback of $\mathcal{L}_{\psi_q(\alpha b)}$ from the base, so projection formula does not create relative vanishing cycles.

Theorem 10.9 (Uniform normal form on every ordered boundary stratum). *After a finite separable extension, write the distinct roots of the squarefree polynomial D as $d_1, \dots, d_k$. On*

$$U = (\mathbb{A}^1 - \{d_1, \dots, d_k\})^n$$

consider the ordered-root map

$$\rho_{\text{ord}} : U \longrightarrow T = (\mathbb{G}_m)^k \times \mathbb{A}_b^1, \quad (x_i)_i \longmapsto \left(\left(\prod_i (d_j - x_i) \right)_{1 \leq j \leq k}, -\sum_i x_i \right),$$

and let $\bar{\Gamma}_{\text{ord}}$ be the reduced closure of its graph in $(\mathbb{P}^1)^n \times T$. At a geometric boundary point, let r be the number of roots equal to infinity and let e_j be the number equal to d_j . Then

$$r \geq 2, \quad e_j \geq 1 \quad (1 \leq j \leq k).$$

There is an étale neighbourhood of the point together with a T -étale morphism of reduced pairs to $(Z, Z^\circ) \times T$, where (Z, Z°) is fixed over the ground field. More explicitly, put

$$P = \prod_{i=2}^r y_i, \quad Q = \sum_{i=2}^r \prod_{\substack{2 \leq \ell \leq r \\ \ell \neq i}} y_\ell.$$

Up to an affine factor of dimension $n - r - \sum_j e_j$, one may take

$$(10.11) \quad Z = V \left(P + zQ, \prod_{\nu=1}^{e_j} v_{j,\nu} - zP \quad (1 \leq j \leq k) \right),$$

and Z° is obtained by inverting

$$z \prod_{i=2}^r y_i \prod_{j=1}^k \prod_{\nu=1}^{e_j} v_{j,\nu}.$$

Consequently, extension by zero from the open ordered graph is universally locally acyclic over T at every boundary point.

Proof. Choose $y_i = x_i^{-1}$ for the r roots at infinity, choose $u_{j,\nu} = d_j - x_{j,\nu}$ for the e_j roots at d_j , and denote all remaining finite-root coordinates by w . Set $Y = \prod_{i=1}^r y_i = y_1 P$. Up to multiplication by units, the local equations of the graph closure are

$$(10.12) \quad BY + \sum_{i=1}^r \frac{Y}{y_i} = 0,$$

$$(10.13) \quad V_j(y, u, w) \prod_{\nu=1}^{e_j} u_{j,\nu} - a_j Y = 0 \quad (1 \leq j \leq k),$$

where $B = b + \sum_{\text{finite } i} x_i$ is regular and every V_j is a unit. These equations imply $e_j \geq 1$ at every boundary point: if no root approached d_j , the second equation would restrict to a nonzero unit. If $r = 1$, the first equation restricts to $1 = 0$ at $y_1 = 0$, so $r \geq 2$.

Use the coordinate

$$z = \frac{y_1}{1 + By_1}, \quad y_1 = \frac{z}{1 - Bz}.$$

It is an isomorphism after shrinking at the boundary, and (10.12) becomes, up to a unit, $P + zQ = 0$. Let V'_j denote V_j after this substitution and set

$$v_{j,1} = a_j^{-1}(1 - Bz)V'_j u_{j,1}, \quad v_{j,\nu} = u_{j,\nu} \quad (\nu > 1).$$

At the boundary the Jacobian of this simultaneous change in the variables $u_{j,1}$ is diagonal with unit diagonal entries: every cross derivative contains the vanishing factor $u_{j,1}$. The change is therefore étale. Equation (10.13) becomes exactly

$$\prod_{\nu=1}^{e_j} v_{j,\nu} = zP.$$

The equations and the vanishing loci defining the complement of the open graph are now independent of (a, b) and of w .

For completeness, these equations give the scheme-theoretic reduced closure. The polynomial $P + zQ$ is geometrically irreducible because it is linear in z , $\gcd(P, Q) = 1$, and Gauss's lemma applies. Thus

$$A_0 = \overline{\mathbb{F}_q}[z, y_2, \dots, y_r]/(P + zQ)$$

is a domain. Moreover $zP \neq 0$ in A_0 , as is also clear from the injection into $\overline{\mathbb{F}_q}[y_2, \dots, y_r, 1/Q]$ given by $z \mapsto -P/Q$. If A is a domain and $0 \neq g \in A$, then

$$A[t_1, \dots, t_e]/(t_1 \cdots t_e - g)$$

is a domain: the unit leading coefficient gives an A -torsion-free Gröbner normal form, while after passage to $K = \text{Frac}(A)$ the quotient is the Laurent domain $K[t_1^{\pm 1}, \dots, t_{e-1}^{\pm 1}]$. Applying this observation to the k groups of v -variables proves that Z is geometrically integral. Its torus open is nonempty: choose general nonzero y_i with $Q \neq 0$, put $z = -P/Q$, and in each v -group choose all but one coordinate equal to 1. It is therefore dense, so Z is exactly the reduced closure of the open graph.

The displayed coordinate construction gives the asserted T -étale morphism to the fixed pair $(Z, Z^\circ) \times T$. Extension by zero is universally locally acyclic over T , because this is étale local and holds for the fixed product pair. Since the assertion is geometric, it descends from the splitting field of D . $\square$

Corollary 10.10 (No ray-boundary vanishing cycles). *Let $\mathcal{L}$ be any lisse sheaf on the ray base; in particular one may take a finite Kummer sheaf tensored with $\mathcal{L}_{\psi_q(\alpha b)}$. Write $\pi : \overline{\Gamma}_{\text{ord}} \rightarrow \mathbb{R}_D$ for the projection. At every ordered boundary point and for every morphism from a trait $\gamma : S \rightarrow \mathbb{R}_D$,*

$$R\Phi_\gamma(j_! \overline{\mathbb{Q}}_\ell \otimes \pi^* \mathcal{L}) = 0.$$

In particular the relative Swan and Artin conductors vanish there. The conclusion is preserved under finite direct sums, shifts, cones, and $\overline{\mathbb{Q}}_\ell$ -linear direct summands. It also holds after proper pushforward by a map whose entire inverse image of the target neighbourhood lies

in these boundary ULA charts; this applies in particular to the finite ordered-to-unordered quotient and its symmetric-group isotypic summands near the boundary.

Proof. The first assertion follows from [theorem 10.9](#), since tensoring by a lisse sheaf pulled back from the base preserves universal local acyclicity. Proper pushforward preserves universal local acyclicity under the stated neighbourhood hypothesis, and the class of universally locally acyclic complexes is thick, which proves the last assertions. $\square$

The old finite-modulus proof cannot simply be cited without this calculation. Its first literal mismatch is [\[21, Lemma 3.1\]](#): after adding the coefficient equation, the boundary is no longer visibly the old product $Z \times \mathbb{G}_D$. The fractional-linear coordinate z and the simultaneous rescaling of one collision coordinate over every zero of D recover a product on all deeper mixed strata as well. Thus the universal-local-acyclicity and vertical-singular-support arguments of [\[21, Lemmas 3.7 and 3.8\]](#) apply formally to the ordered squarefree $D\infty^2$ boundary and to the finite factorization summands just described. What they do not supply is a uniform bound for the unsigned Betti numbers after Fourier transform.

10.5. The Artin–Schreier collapse. There is a second reduction which is invisible before taking a nonzero Fourier stalk. Put

$$C_D = \mathbb{A}^1 - V(D), \quad \mathcal{A}_\alpha = \mathcal{L}_{\psi_q(-\alpha x)}|_{C_D} \quad (\alpha \neq 0),$$

and, after splitting D with roots $d_1, \dots, d_k$, write

$$\pi_D : C_D^n \longrightarrow (\mathbb{G}_m)^k, \quad (x_i)_i \longmapsto \left(\prod_{i \leq j \leq k} (d_j - x_i) \right).$$

For a $\overline{\mathbb{Q}}_\ell$ -representation ρ of S_n , define

$$(10.14) \quad \mathcal{P}_{\rho, \alpha} := \left(R\pi_{D,!} \mathcal{A}_\alpha^{\boxtimes n} \otimes \rho \right)^{S_n}.$$

Its trace at a is

$$\sum_{\substack{f \in \mathbf{M}_n(\mathbb{F}_q) \\ f \equiv a \pmod{D}}} F_\rho(f) \psi_q(\alpha c_1(f)),$$

where F_ρ is Sawin’s factorization function. Thus the von Mangoldt twist is the alternating sum of (10.14) for $\rho = \bigwedge^i \text{std}$.

Lemma 10.11 (One-variable Artin–Schreier cohomology). *Geometrically,*

$$H_c^i(C_D, \mathcal{A}_\alpha) = 0 \quad (i \neq 1), \quad \dim H_c^1(C_D, \mathcal{A}_\alpha) = k.$$

Consequently

$$(10.15) \quad R\Gamma_c((\mathbb{G}_m)^k, \mathcal{P}_{\rho, \alpha}) \simeq \left(H_c^1(C_D, \mathcal{A}_\alpha)^{\otimes n} \otimes \text{sgn} \otimes \rho \right)^{S_n}[-n].$$

In particular the left side is concentrated in ordinary degree n .

Proof. The sheaf $\mathcal{A}_\alpha$ is geometrically nonconstant and has Swan conductor one at infinity. Hence $H_c^0 = H_c^2 = 0$. The Grothendieck–Ogg–Shafarevich formula gives

$$\chi_c(C_D, \mathcal{A}_\alpha) = \chi_c(C_D, \overline{\mathbb{Q}}_\ell) - \text{Swan}_\infty(\mathcal{A}_\alpha) = (1 - k) - 1 = -k,$$

which proves the first assertion. Proper base change, Künneth, and exactness of S_n -invariants give (10.15). The sign occurs because all n Künneth factors have odd cohomological degree. $\square$

Theorem 10.12 (Mellin perversity at fixed frequency). *For every $\overline{\mathbb{Q}}_\ell$ -representation ρ of S_n and every $\alpha \neq 0$, the complex*

$$\mathcal{P}_{\rho, \alpha}[n] = \left(R\pi_{D,!} \mathcal{A}_\alpha^{\boxtimes n} \otimes \rho \right)^{S_n}[n]$$

is perverse on $(\mathbb{G}_m)^k$. This holds in arbitrary characteristic.

Proof. Put $T = (\mathbb{G}_m)^k$ and

$$g : C_D \longrightarrow T, \quad x \longmapsto (d_1 - x, \dots, d_k - x).$$

The map g is a closed immersion: its image is cut out by $z_j - z_1 = d_j - d_1$. Hence

$$\mathcal{M}_\alpha := g_* \mathcal{A}_\alpha[1]$$

is perverse on T .

Let $\mathfrak{M}_*$ be the universal Mellin transform of Gabber–Loeser [10], with coefficient ring on their character scheme $\mathcal{C}(T)$. The derived fiber of $\mathfrak{M}_*(\mathcal{M}_\alpha)$ at a geometric tame character χ is

$$R\Gamma(C_D, \mathcal{A}_\alpha \otimes g^* \mathcal{L}_\chi)[1].$$

The rank-one twist is still wildly nontrivial at infinity with Swan conductor one. Its H^0 vanishes, while an affine curve has no ordinary cohomology above degree one. The Grothendieck–Ogg–Shafarevich formula, together with equality of ordinary and compactly supported Euler characteristics, therefore gives

$$H^i(C_D, \mathcal{A}_\alpha \otimes g^* \mathcal{L}_\chi) = 0 \quad (i \neq 1), \quad \dim H^1 = k.$$

The universal Mellin complex is perfect. Since all of its derived fibers have amplitude zero and the same dimension, derived Nakayama and cohomology-and-base-change show that it is a finite locally free $\mathcal{O}_{\mathcal{C}(T)}$ -module of rank k , placed in degree zero.

The map π_D is multiplication on T after applying g to each coordinate. Thus

$$R\pi_{D,*} \mathcal{A}_\alpha^{\boxtimes n}[n] \simeq \mathcal{M}_\alpha * \dots * \mathcal{M}_\alpha.$$

Universal Mellin transform carries $*$ to derived tensor product. Its transform is consequently the n -fold tensor power of the rank- k vector bundle above, again in degree zero. The Gabber–Loeser recognition theorem [10, Theorems 3.4.1 and 3.4.7] says that a constructible complex on a torus is perverse if and only if its universal $\mathfrak{M}_*$ -transform is concentrated in degree zero. It follows that

$$R\pi_{D,*} \mathcal{A}_\alpha^{\boxtimes n}[n]$$

is perverse.

Finally, Verdier duality gives, up to the harmless Tate twist (n) ,

$$\mathbb{D}_T \left(R\pi_{D,*} \mathcal{A}_\alpha^{\boxtimes n}[n] \right) \simeq R\pi_{D,!} \mathcal{A}_\alpha^{\boxtimes n}n.$$

The perverse heart is Verdier self-dual. Tensoring with ρ and taking S_n -invariants is an idempotent direct-summand operation over $\overline{\mathbb{Q}}_\ell$, so it preserves perversity. $\square$

Theorem 10.13 (Fixed-frequency boundary cleanliness). *Let X_α be the reduced closure of the graph of π_D in $(\mathbb{P}^1)^n \times T$, and let $u : C_D^n \hookrightarrow X_\alpha$ be the open immersion. Then*

$$u_! \mathcal{A}_\alpha^{\boxtimes n}$$

is universally locally acyclic over T in a neighbourhood of the entire boundary $X_\alpha \setminus C_D^n$.

Proof. Work after a finite separable extension splitting D . At a boundary point let $r \geq 1$ roots tend to infinity, let e_j roots tend to d_j , and write

$$y_i = x_i^{-1}, \quad u_{j,\nu} = d_j - x_{j,\nu}, \quad Y = \prod_{i=1}^r y_i.$$

The j th graph equation is

$$U_j(y, u, w) \prod_{\nu=1}^{e_j} u_{j,\nu} = a_j Y,$$

where U_j is a unit. It follows that every $e_j \geq 1$. Replacing one coordinate in each block by

$$v_{j,1} = a_j^{-1} U_j u_{j,1}, \quad v_{j,\nu} = u_{j,\nu} \quad (\nu > 1)$$

is an étale change of variables: at the boundary its Jacobian in the chosen coordinates is diagonal with unit diagonal. The equations become the base-independent normal form

$$(10.16) \quad \prod_{\nu=1}^{e_j} v_{j,\nu} = Y \quad (1 \leq j \leq k).$$

The Artin–Schreier phase splits as

$$-\alpha \sum_i x_i = -\alpha \sum_{i=1}^r \frac{1}{y_i} + h(y, u, w),$$

with h regular across the boundary. Hence, on the resulting étale chart,

$$u! \mathcal{A}_\alpha^{\boxtimes n} \simeq \mathrm{pr}_Z^* \left((j_Z)! \mathcal{L}_{\psi_q(-\alpha \sum_i 1/y_i)} \right) \otimes \mathcal{L}_{\psi_q(h)},$$

where Z is the fixed scheme (10.16) and j_Z is its torus open. The first factor is pulled back from a fixed pair over the ground field; the second is lisse on the whole chart. A fixed constructible complex is ULA over the ground field, and ULA is preserved by base change, tensoring with a lisse sheaf, and étale pullback. This proves the assertion. Notice that the argument includes $r = 1$: the factor $\mathcal{L}_{\psi_q(-\alpha/y_1)}$ is absolutely wild at $y_1 = 0$, but it is fixed in the relative T -direction. No integer is divided by, so the proof is characteristic-free. $\square$

Corollary 10.14 (Two-degree concentration). *Assume $n \geq k$. For every ρ and every $\alpha \neq 0$,*

$$\mathcal{P}_{\rho,\alpha} \text{ is concentrated in ordinary degrees } n - k \text{ and } n + 1 - k.$$

Proof. The lower bound follows from theorem 10.12, since the base torus has dimension k . For the upper bound, apply the vanishing-cycle argument of [21, Lemma 4.6]. Its boundary input is exactly theorem 10.13. Away from a boundary neighbourhood, the non-ULA support is both proper and affine over T , hence finite. The fibers of π_D have dimension $n - k$, so semiperversity of vanishing cycles shows that specialization from the geometric generic point can add only ordinary degree $n + 1 - k$. At the generic point perversity leaves only degree $n - k$. $\square$

Proposition 10.15 (Long-row annihilation). *Let ρ_λ be the irreducible representation indexed by a partition $\lambda \vdash n$. If*

$$\lambda_1 \geq k + 1,$$

then, for every $\alpha \neq 0$,

$$\mathcal{P}_{\rho_\lambda,\alpha} = 0.$$

Equivalently, the nonzero infinity Fourier mode kills every factorization isotype having a row of length at least $k + 1$.

Proof. Write $\lambda = (n_1, \dots, n_s)$ with $n_1 \geq k + 1$. The Young permutation representation

$$\mathrm{Ind}_{S_{n_1} \times \dots \times S_{n_s}}^{S_n} \overline{\mathbb{Q}}_\ell$$

contains ρ_λ as a direct summand. It therefore suffices, by Frobenius reciprocity and the finite ordered-to-symmetric quotient, to work on

$$C_D^{(n_1)} \times \dots \times C_D^{(n_s)}.$$

Regard a point as a tuple of monic polynomials $(F_1, \dots, F_s)$ prime to D . The augmented multiplication map is

$$(F_i)_i \longmapsto \left(\prod_i F_i \bmod D, \sum_i c_1(F_i) \right) \in \mathbb{G}_D \times \mathbb{A}_b^1.$$

There is an isomorphism

$$C_D^{(n_1)} \times \dots \times C_D^{(n_s)} \simeq \mathbb{G}_D \times \mathbb{A}_b^1 \times \mathbb{A}^{n_1-k-1} \times C_D^{(n_2)} \times \dots \times C_D^{(n_s)}.$$

Indeed, b determines $c_1(F_1)$, the coefficients $c_2(F_1), \dots, c_{n_1-k}(F_1)$ are free, and the remaining k coefficients are determined uniquely by the invertible residue

$$F_1 \bmod D = a \left(\prod_{i=2}^s F_i \bmod D \right)^{-1}.$$

Polynomial division shows that the inverse is regular. This is the coefficient-enhanced version of [21, Lemma 4.2].

The augmented pushforward is therefore a constant complex on the b -line. Its Fourier stalk at $\alpha \neq 0$ contains the factor

$$R\Gamma_c(\mathbb{A}_b^1, \mathcal{L}_{\psi_q(\alpha b)}) = 0.$$

The induced object vanishes, and so does its direct summand corresponding to ρ_λ . $\square$

Remark 10.16 (Why the threshold moves by one). For the finite modulus alone, a row of length at least k produces a geometrically constant main term. After adjoining c_1 , a row of length at least $k+1$ again produces a constant family, but its nonzero Fourier transform is zero. Rows of length exactly k no longer belong to the main term: they survive in middle degree. This is the representation-theoretic counterpart of [theorem 10.5](#), where the last surviving characteristic-cycle components have $s(w) = k$.

10.6. The conic polar term and the transverse generic rank. We now separate two numbers which play different geometric roles. For a weight function w put

$$s(w) = \sum_{j \geq 1} w(j), \quad S_w = \prod_{j \geq 1} S_j^{w(j)}, \quad M_\rho(w) = \dim \left((\rho \otimes \text{sgn})^{S_w} \right).$$

If $s(w) = r$, set

$$(10.17) \quad d_w^{(r)} := \frac{r!}{\prod_{j \geq 1} w(j)!} \prod_{j \geq 1} j^{w(j)}$$

and define

$$(10.18) \quad C_r(\rho) := \sum_{\substack{w: \sum_j j w(j) = n \\ s(w) = r}} M_\rho(w) d_w^{(r)}.$$

Equivalently, if

$$V_{\rho, r} = \left((\overline{\mathbb{Q}}_\ell^T)^{\otimes n} \otimes \rho \otimes \text{sgn} \right)^{S_n},$$

then the orbit grouping used in [21, Lemma 6.13] gives

$$(10.19) \quad C_r(\rho) = \partial_{\lambda_1} \cdots \partial_{\lambda_r} \text{tr}(\text{diag}(\lambda_1, \dots, \lambda_r), V_{\rho, r})|_{\lambda_1 = \dots = \lambda_r = 1}.$$

We put $C_0(\rho) = 0$ when $n > 0$.

Theorem 10.17 (Fixed-frequency conic polar bound). *Let*

$$K_{\rho, \alpha} := \mathcal{P}_{\rho, \alpha}[n].$$

For every characteristic and every $\alpha \neq 0$, its characteristic cycle on $T = (\mathbb{G}_m)^k$ has the form

$$(10.20) \quad CC(K_{\rho, \alpha}) = r_{\rho, \alpha} [T_T^* T] + \sum_{\substack{w \\ s(w) < k}} M_\rho(w) [B'_w],$$

where $r_{\rho, \alpha}$ is the geometric generic rank and B'_w is Sawin's homogeneous proper-pushforward cycle for the finite map π_D . Consequently, at every geometric point a of T ,

$$(10.21) \quad \dim \mathcal{H}^{n+1-k}(\mathcal{P}_{\rho, \alpha})_a \leq C_{k-1}(\rho).$$

Proof. On the affine symmetric-power space the Artin–Schreier factor $\mathcal{L}_{\psi_q(-\alpha c_1)}$ is lisse. Tensoring by it does not change the characteristic cycle. Sawin’s symmetric-power formula [20, Theorem 2.10] therefore gives the same interior cycles B_w with multiplicities $M_\rho(w)$ as for the constant rank-one local system.

For the map to the fixed k -dimensional torus T , the transpose condition is the homogeneous condition with no infinity-frequency coordinate. The proper-pushforward argument of [21, Lemmas 6.10–6.11] sends $s(w) < k$ to B'_w and sends $s(w) \geq k$ to the zero section. Its properness hypothesis holds here: a point in the graph boundary has the $k + 1$ distinct coordinate values

$$d_1, \dots, d_k, \infty,$$

whereas a point of the w -stratum has only $s(w) < k$ distinct support values. Finally, [theorem 10.13](#) says that a boundary characteristic covector pulled back from T must be zero. Boundary components can therefore change only the coefficient of the zero section. More precisely, the proper-pushforward formula first gives (10.20) modulo the zero section. Every B'_w in the displayed sum lies over a proper closed subset of T , since $s(w) < k$; restriction to the geometric generic point therefore identifies the remaining zero-section coefficient with the generic rank $r_{\rho, \alpha}$. This proves (10.20).

Projectivization removes the zero section. In the polar pushforward, components with $s(w) < k - 1$ have image of dimension below $k - 1$ and vanish. If $s(w) = k - 1$, the pullback of a hyperplane has class $\sum_j j H_j$, so its degree is bounded by

$$\left(\sum_j j H_j \right)^{k-1} = \frac{(k-1)!}{\prod_j w(j)!} \prod_j j^{w(j)} = d_w^{(k-1)}$$

on the component indexed by w . The positive-characteristic Massey inequality [19, Theorem 1.4], exactly as applied in [21, Proposition 6.14], bounds the degree $1 - k$ stalk of the perverse sheaf $K_{\rho, \alpha}$ by this polar multiplicity. After undoing the shift $[n]$, this is (10.21). $\square$

Lemma 10.18 (Lisse local terms retain intersection multiplicity). *Let X be smooth over an algebraically closed field in a neighbourhood of an isolated fixed point x of an automorphism s . Let $\mathcal{L}$ be lisse near x , with a cohomological correspondence $u : s^* \mathcal{L} \rightarrow \mathcal{L}$. Then the true Lefschetz–Verdier local term is*

$$(10.22) \quad LT_x(s, \mathcal{L}, u) = i_x(\Gamma_s, \Delta; X \times X) \operatorname{Tr}(u_x | \mathcal{L}_x).$$

If the fixed scheme is zero-dimensional near x , then

$$i_x(\Gamma_s, \Delta; X \times X) = \operatorname{length} \mathcal{O}_{\operatorname{Fix}(s), x}.$$

No assumption on the order of s relative to the characteristic is needed.

Proof. For the constant sheaf this is the local cycle-class calculation in the Lefschetz–Verdier formula [11, Exposé III, Section 4.3]; see also [14, Exposé III B, Section 6.14]. On a strict henselian neighbourhood of x , a lisse complex is a constant perfect complex. The Verdier evaluation contracts the coefficient endomorphism to its trace, while the geometric factor is the constant-sheaf intersection class; this gives (10.22). Finally, Γ_s and Δ are regular of complementary codimension in the smooth scheme $X \times X$. On $\Gamma_s \simeq X$, the diagonal equations form a system of parameters. Smoothness makes the local ring Cohen–Macaulay, so this system is regular and the refined intersection multiplicity is the displayed scheme length. For an explicit curve-level formula with the same nontransverse multiplicity, compare [1, Corollary 1.7.5]. $\square$

Lemma 10.19 (Wild fixed-cycle Lefschetz number). *Let η be the geometric generic point of $T = (\mathbb{G}_m)^k$. If $\sigma \in S_n$ has exactly k cycles of lengths $e_1, \dots, e_k$, then, in every characteristic,*

$$(10.23) \quad \sum_j (-1)^j \operatorname{Tr} \left(\sigma, H_c^j \left(\pi_D^{-1}(\eta), \mathcal{A}_\alpha^{\boxtimes n} \right) \right) = k! \prod_{i=1}^k e_i.$$

Proof. Scheme-theoretically, the fixed locus is the geometric generic fiber of

$$\varpi_e : C_D^k \longrightarrow T, \quad (y_1, \dots, y_k) \longmapsto \left(\prod_{i=1}^k (d_j - y_i)^{e_i} \right)_{j=1}^k.$$

This map is dominant even if some $p \mid e_i$. In positive characteristic, write $e_i = p^{r_i} m_i$ with $p \nmid m_i$, and put $v_i = y_i^{p^{r_i}}$. After the finite radical map $(y_i) \mapsto (v_i)$, the logarithmic Jacobian has entries

$$\frac{m_i}{d_j^{p^{r_i}} - v_i}.$$

Its determinant is nonzero: at $v_i = d_i^{p^{r_i}}$, only the diagonal term has the full product of the k indicated simple poles. Thus ϖ_e is generically finite. In characteristic zero the same argument uses $r_i = 0$.

In $(\mathbb{P}^1)^k$, its fiber over $a \in T$ is cut out by k Cartier divisors of common class

$$L = \sum_{i=1}^k e_i H_i.$$

There is no common point at infinity. If one coordinate is infinity, the k equations force the other $k - 1$ coordinates to contain all k distinct values $d_1, \dots, d_k$. Consequently the generic fixed scheme is finite, lies in C_D^k , and has length

$$(10.24) \quad L^k = k! \prod_{i=1}^k e_i.$$

This is scheme length, not the number of geometric points.

At a generic fixed point the y_i are pairwise distinct, since every diagonal in C_D^k has dimension $k - 1$ and therefore misses the generic point of T . Choosing one ambient coordinate direction from each cycle gives the Cauchy minor

$$\left(\frac{1}{d_j - y_i} \right)_{j,i}$$

of $d\pi_D$. It is nonsingular, so the ambient ray fiber is smooth there even when the restricted map ϖ_e is inseparable. Moreover a boundary point of the compactified ray fiber uses the $k + 1$ distinct coordinate values $d_1, \dots, d_k, \infty$, whereas a σ -fixed tuple uses at most k values. Hence there are no boundary fixed points. The canonical permutation equivariance of $\mathcal{A}_\alpha^{\boxtimes n}$ has stalk trace 1 at a fixed tuple. Applying the proper Lefschetz–Verdier formula to the graph compactification and using [lemma 10.18](#) and [equation \(10.24\)](#) proves [\(10.23\)](#). $\square$

Remark 10.20 (Where the old tame restriction entered). For a permutation with k cycles, $\sum_i (e_i - 1) = n - k$, so $e_i \leq n - k + 1$. Thus the naive transverse argument already worked under the sharp sufficient condition $p > n - k + 1$, rather than $p > n$. Its missing conjugacy classes were exactly

$$\mathcal{W}_p(n, k) = \left\{ \mu \vdash n : \ell(\mu) = k, \quad p \mid \prod_i \mu_i \right\}.$$

Their number is the coefficient of $x^n y^k$ in

$$\prod_{m \geq 1} (1 - yx^m)^{-1} - \prod_{\substack{m \geq 1 \\ p \nmid m}} (1 - yx^m)^{-1}.$$

The set is empty precisely when $p > n - k + 1$. The preceding lemma fills these columns by retaining their nonreduced intersection lengths; the naive local term would retain only the reduced point count and is therefore insufficient.

Theorem 10.21 (Characteristic-free generic rank). *For every characteristic, every ρ , and every $\alpha \neq 0$,*

$$(10.25) \quad r_{\rho, \alpha} = C_k(\rho).$$

Proof. It is enough to treat irreducible ρ . Let η be the geometric generic point of T and put

$$\chi_\rho = \sum_j (-1)^j \dim \mathcal{H}^j(\mathcal{P}_{\rho, \alpha})_\eta.$$

By [proposition 10.15](#), $\chi_\rho = 0$ if the first row of ρ has length at least $k + 1$. For the same representations Schur–Weyl duality gives $V_{\rho, k} = 0$, and hence $C_k(\rho) = 0$ by [\(10.19\)](#).

Let $\sigma \in S_n$ have $c \leq k$ cycles, of lengths $e_1, \dots, e_c$. A σ -fixed tuple has at most c distinct coordinate values, so the preceding $k + 1$ -point boundary criterion shows that it has no fixed point at the graph boundary. If $c < k$, its fixed locus cannot dominate T and is empty over η . The Lefschetz–Verdier trace is therefore zero in this case, without any condition on the order of σ .

For $c = k$, [lemma 10.19](#) computes the true local term, including its nonreduced multiplicity. We have therefore proved, in every characteristic,

$$(10.26) \quad \sum_\rho \chi_\rho \operatorname{tr}(\sigma, \rho) = \begin{cases} 0, & c < k, \\ k! \prod_i e_i, & c = k. \end{cases}$$

Schur–Weyl duality and [\(10.19\)](#) show that

$$\sum_\rho C_k(\rho) \operatorname{tr}(\sigma, \rho) = \partial_{\lambda_1} \cdots \partial_{\lambda_k} \left[\operatorname{sgn}(\sigma) \prod_{\text{cycles } e} \left(\sum_{j=1}^k \lambda_j^e \right) \right] \Big|_{\lambda=1}.$$

It vanishes for $c < k$ and equals $(-1)^{n-k} k! \prod_i e_i$ for $c = k$. Thus

$$\tilde{\chi}_\rho := (-1)^{n-k} C_k(\rho)$$

satisfies [\(10.26\)](#); by the long-row vanishing above, it also agrees with χ_ρ on every partition whose first row has length at least $k + 1$. The character-table uniqueness argument of [\[21, Lemma 5.4\]](#), with $m = k + 1$, now gives

$$\chi_\rho = (-1)^{n-k} C_k(\rho).$$

By [corollary 10.14](#), the generic stalk is concentrated in degree $n - k$. Hence its rank is $C_k(\rho)$. $\square$

Corollary 10.22 (All-characteristic nonzero-frequency ray estimate). *Let $n \geq k \geq 1$ and $\alpha \neq 0$. In every characteristic, put*

$$(10.27) \quad G_{n,k} := 2^{k-1} \binom{n+k-1}{2k-1},$$

$$(10.28) \quad P_{n,k} := \begin{cases} 0, & k = 1, \\ 2^{k-2} \binom{n+k-2}{2k-3}, & k \geq 2. \end{cases}$$

Then, uniformly in squarefree D of degree k and $a \in A_D^\times$,

$$(10.29) \quad |S_\alpha(n; D, a)| \leq (G_{n,k} + \sqrt{q} P_{n,k}) q^{(n-k)/2}.$$

Proof. For the unsigned von Mangoldt representation

$$\rho_\Lambda = \bigoplus_{i=0}^{n-1} \bigwedge^i \operatorname{std},$$

Sawin’s hook-character identity [\[21, Lemma 8.14\]](#) is

$$\operatorname{tr}(\operatorname{diag}(\lambda_1, \dots, \lambda_r), V_{\rho_\Lambda, r}) = \frac{1}{2} [u^n] \prod_{i=1}^r \frac{1 + u\lambda_i}{1 - u\lambda_i}.$$

For $r \geq 1$, differentiating every variable and setting $\lambda_i = 1$ gives

$$(10.30) \quad C_r(\rho_\Lambda) = 2^{r-1}[u^{n-r}](1-u)^{-2r} = 2^{r-1} \binom{n+r-1}{2r-1}.$$

Thus $C_k(\rho_\Lambda) = G_{n,k}$; for $k \geq 2$ the same formula gives $C_{k-1}(\rho_\Lambda) = P_{n,k}$, while for $k = 1$ this equality is the separate convention $C_0 = P_{n,1} = 0$. Notice that the second number uses the $(k-1)$ -colour character; leaving one variable undifferentiated in the k -colour character would incorrectly mix the generic $s(w) = k$ layer into the polar term. For example, $P_{n,2} = n$.

The lowest stalk degree of a perverse sheaf on the normal torus cannot have rank larger than at the generic point. Hence [theorem 10.21](#) bounds the degree $n-k$ part by $G_{n,k}$, while [theorem 10.17](#) bounds the degree $n+1-k$ part by $P_{n,k}$. Deligne's weight bound contributes respectively $q^{(n-k)/2}$ and $q^{(n+1-k)/2}$. Summing the surviving hooks in the alternating factorization formula for Λ proves [\(10.29\)](#). $\square$

Proposition 10.23 (The wild columns are genuinely necessary). *Assume $k \geq 2$ and $p \leq n-k+1$, and let*

$$\mu = (p, n-p-k+2, 1^{k-2})$$

be reordered as a partition. The honest representation

$$W_{n,k,p} = \text{sgn} \otimes \text{Ind}_{S_{\mu_1} \times \cdots \times S_{\mu_k}}^{S_n} \mathbf{1}$$

has the following properties:

- (1) *every constituent S^λ satisfies $\lambda_1 \leq k$;*
- (2) *its character vanishes on every p -regular permutation with at most k cycles; and*
- (3)

$$\dim \text{Hom}_{S_n}(\rho_\Lambda, W_{n,k,p}) = 2^{k-1}.$$

Consequently, long-row annihilation, positivity, and all tame character equations alone do not imply any upper bound for the von Mangoldt generic rank.

Proof. The induced representation is the permutation representation on μ -tabloids. A permutation with fewer than k cycles fixes no tabloid with k nonempty rows. If it has exactly k cycles, a fixed tabloid forces its cycle-length multiset to be μ , which is not p -regular. This proves the second assertion.

Young's rule writes the permutation representation with multiplicities $K_{\nu\mu}$. A semistandard tableau using k symbols has at most k rows; the sign twist transposes ν , proving the first assertion. For a hook,

$$K_{(n-r, 1^r), \mu} = \binom{k-1}{r} \quad (0 \leq r \leq k-1) :$$

choose the r symbols below the upper-left box, after which the first row is forced. Summing over the hooks in ρ_Λ gives $\sum_r \binom{k-1}{r} = 2^{k-1}$. Adding arbitrarily many copies of $W_{n,k,p}$ preserves every item of the tame data while increasing the hook multiplicity, proving the final claim. $\square$

Theorem 10.24 (Degree-one modulus is exactly Kloosterman). *Let $D = T - d$, $a \in \mathbb{F}_q^\times$, and $\alpha \in \mathbb{F}_q^\times$. Define*

$$\text{Kl}_n(b; \psi_q) = \sum_{\substack{u_1, \dots, u_n \in \mathbb{F}_q^\times \\ u_1 \cdots u_n = b}} \psi_q(u_1 + \cdots + u_n).$$

Then, in every characteristic,

$$(10.31) \quad S_\alpha(n; T - d, a) = (-1)^{n-1} \psi_q(-\alpha nd) \text{Kl}_n(\alpha^n a; \psi_q).$$

Consequently,

$$(10.32) \quad |S_\alpha(n; T - d, a)| \leq nq^{(n-1)/2}.$$

Proof. Translate $g(U) = f(U + d)$. Then $g(0) = a$ and $c_1(g) = c_1(f) + nd$. For a multiplicative character χ of $\mathbb{F}_q^\times$, put

$$G_\alpha(\chi) = \sum_{x \in \mathbb{F}_q^\times} \chi(x) \psi_q(\alpha x)$$

and consider the Euler product

$$L_\chi(u) = \sum_{\substack{g \text{ monic} \\ g(0) \neq 0}} \chi(g(0)) \psi_q(\alpha c_1(g)) u^{\deg g}.$$

Constant terms multiply and first coefficients add under multiplication of monic polynomials. The degree- m coefficient vanishes for $m \geq 2$ by summing the free coefficient $c_1(g)$, while the degree-one coefficient is $G_\alpha(\chi)$. Hence

$$L_\chi(u) = 1 + G_\alpha(\chi)u.$$

Taking the logarithmic derivative of its prime-polynomial Euler product gives

$$\sum_{\deg g=n} \Lambda(g) \chi(g(0)) \psi_q(\alpha c_1(g)) = (-1)^{n-1} G_\alpha(\chi)^n.$$

Multiplicative Fourier inversion at $g(0) = a$, together with

$$\sum_{b \in \mathbb{F}_q^\times} \chi(b) \text{Kl}_n(b; \psi_q) = G_1(\chi)^n$$

and $G_\alpha(\chi) = \chi(\alpha^{-1}) G_1(\chi)$, proves (10.31). The hyper-Kloosterman sheaf has rank n and weight $n - 1$, so Deligne’s bound gives (10.32); see [15, Chapters 4 and 11]. $\square$

Remark 10.25 (What the local-intersection step adds). The conic characteristic-cycle calculation [theorem 10.17](#) controls the special-stalk polar term but cannot see the zero-section coefficient. Likewise, relative ULA alone cannot exclude a constant lisse summand. The missing input is precisely the true local term in [lemma 10.18](#): in the wild case it keeps the nilpotent length discarded by the naive stalk trace. The identity (10.31) independently calibrates this phenomenon: even when $p \mid n$, the rank remains n , rather than dropping to the prime-to- p part of n .

10.7. Unsigned cohomology of the fixed boundary model. The product statement in [theorem 10.9](#) can be made quantitative locally. This does not yet count the global incidence of the strata, but it proves that a single deep mixed singularity is not the source of an uncontrolled Betti number.

Proposition 10.26 (Exact open boundary model). *Use the notation of (10.11), put*

$$m = r - 1, \quad E = \sum_{j=1}^k e_j, \quad s = E - k,$$

and let

$$H_m = \{(t_1, \dots, t_m) \in \mathbb{G}_m^m : t_1 + \dots + t_m = -1\}.$$

Then

$$Z^\circ \simeq \mathbb{G}_m^{s+1} \times H_m.$$

In particular, with $d = \dim Z = m + s$,

$$\sum_i \dim H_c^i(Z, j; \overline{\mathbb{Q}}_\ell) = 2^{s+1} (2^m - 1) \leq 3^d.$$

All these groups are mixed Tate; more precisely

$$P(H_m; t) = (1 + t)^m - t^m.$$

Proof. On Z° , put $t_i = z/y_i$ for $1 \leq i \leq m$. Dividing $P + zQ = 0$ by P gives $1 + \sum_i t_i = 0$. Conversely, $z \in \mathbb{G}_m$ and $t \in H_m$ determine $y_i = z/t_i$ and

$$zP = \frac{z^{m+1}}{\prod_i t_i}.$$

In each v -block, choose $e_j - 1$ coordinates freely and use $\prod_\nu v_{j,\nu} = zP$ to determine the last. This gives the asserted isomorphism. The variety H_m is the complement of m hyperplanes in general position in $\mathbb{A}^{m-1}$, so

$$H^i(H_m, \overline{\mathbb{Q}}_\ell) \simeq \overline{\mathbb{Q}}_\ell(-i)^{\binom{m}{i}} \quad (0 \leq i \leq m-1).$$

Künneth and Poincaré duality give the formula. Finally, $2^{s+1}(2^m - 1) \leq 3^{m+s}$. $\square$

Proposition 10.27 (Boundary and total Betti bounds). *Let $B = (Z \setminus Z^\circ)_{\text{red}}$. Then*

$$\sum_i \dim H_c^i(B, \overline{\mathbb{Q}}_\ell) < 3^d, \quad \sum_i \dim H_c^i(Z, \overline{\mathbb{Q}}_\ell) \leq 2 \cdot 3^d.$$

Restoring the affine factor omitted in [theorem 10.9](#) does not alter total Betti number, and $d \leq n - k - 1$. The bounds hold in arbitrary characteristic and do not acquire a factorial factor after a finite symmetric quotient or passage to an isotypic direct summand.

Proof. Write

$$X = V(P + zQ) \subset \mathbb{A}^{m+1}, \quad X_0 = (V_X(zP))_{\text{red}}, \quad F_e = V(v_1 \cdots v_e) \subset \mathbb{A}^e.$$

The defining equations give the product decomposition

$$B \simeq X_0 \times \prod_{j=1}^k F_{e_j}.$$

If exactly one y_i vanishes on X_0 , then $z = 0$; if $c \geq 2$ of them vanish, z is free. The resulting torus stratification gives

$$\beta_c(X_0) \leq 3^m - 2^m.$$

The open–closed sequence for $F_e = \mathbb{A}^e \setminus \mathbb{G}_m^e$ gives exactly

$$\beta_c(F_e) = 2^e - 1 \leq 3^{e-1}.$$

Künneth now yields

$$\beta_c(B) \leq (3^m - 2^m) \prod_j (2^{e_j} - 1) < 3^{m+E-k} = 3^d.$$

The open–closed triangle and [proposition 10.26](#) give the bound for Z . Taking invariants under a finite group is exact over $\overline{\mathbb{Q}}_\ell$, and a projector-defined summand cannot have larger cohomology, proving the last assertion. $\square$

10.8. Numerical payoff of the wild local term. Both the polar multiplicity and its phase-sensitive generic-rank companion are now explicit in every characteristic. For $1/2 < \vartheta < 1$, put

$$(10.33) \quad A_\vartheta := 2^{\vartheta/(1-\vartheta)} \frac{e(1+\vartheta)}{1-\vartheta}.$$

This is the exponential constant in Sawin’s von Mangoldt calculation [[21](#), Proposition 8.16].

For $k \geq 2$, the two exact constants satisfy

$$(10.34) \quad \frac{P_{n,k}}{G_{n,k}} = \frac{(2k-1)(k-1)}{(n+k-1)(n-k+1)}.$$

If $N = n - k$ and $k \leq \vartheta n$, then

$$(10.35) \quad G_{n,k} \leq \frac{1}{2} A_\vartheta^N, \quad P_{n,k} \leq \frac{2\vartheta^2}{1-\vartheta} G_{n,k}.$$

Thus the added condition at infinity introduces neither a new exponential constant nor a hidden Euler-tail factor.

Theorem 10.28 (All-characteristic bounded-additive-sieve range). *Let $1/2 < \vartheta < 1$. Then $\text{BAS}_1(\theta, \delta)$ holds for every $\theta < \vartheta$ and every*

$$0 < \delta < \frac{1}{2} - \log_q A_\vartheta.$$

In particular the exponent crosses $1/2$ whenever $q > A_\vartheta^2$. As $\vartheta \downarrow 1/2$, this condition tends to $q > 36e^2$; equivalently $q > \lfloor 36e^2 \rfloor = 266$.

Proof. The all-characteristic pointwise bound is [corollary 10.22](#). Since

$$G_{n,k} = 2^{k-1} \binom{N+2k-1}{N},$$

the inequality $\binom{a}{b} \leq (ea/b)^b$ gives the first estimate in [\(10.35\)](#); the second follows directly from [\(10.34\)](#). Hence

$$|S_\alpha(n; D, a)| \ll_{q,\vartheta} A_\vartheta^{n-k} q^{(n-k)/2} = q^{(n-k)(1-\delta_0)}, \quad \delta_0 = \frac{1}{2} - \log_q A_\vartheta.$$

The strict inequalities $\theta < \vartheta$ and $\delta < \delta_0$ absorb the finitely many initial degrees. $\square$

Corollary 10.29 (A fixed-field range beyond one half). *If $q > 266$, then there exist $\theta > 1/2$ and $\delta > 0$ for which $\text{BAS}_1(\theta, \delta)$ holds in every characteristic. Equivalently, the nonzero Fourier modes in the small ray column [\(9.2\)](#) have a uniform power saving throughout a divisor-degree range strictly larger than one half.*

Proof. Choose $\vartheta > 1/2$ sufficiently close to $1/2$ that $q > A_\vartheta^2$, then choose $1/2 < \theta < \vartheta$ and $0 < \delta < 1/2 - \log_q A_\vartheta$. $\square$

Remark 10.30 (Status after the calculation). [Theorem 10.5](#) is the first new algebraic information obtained in this step: the nonzero Fourier mode removes every interior characteristic-cycle component with more than k distinct support points, in arbitrary characteristic. The classical RH estimate [proposition 10.3](#) and the Fourier dictionary are not new. [Theorem 10.9](#) and [corollary 10.10](#) add the second algebraic point: for squarefree $D\infty^2$, every ordered boundary stratum is clean in the ray directions, including all deep mixed collisions. The Mellin theorem and [corollary 10.14](#) and [theorem 10.17](#) add the third: perversity, two-degree concentration, and the full special-stalk polar bound are unconditional. The generic rank is computed in [theorem 10.21](#) in every characteristic: the classical SGA 5 local-intersection formula retains the inseparable fixed-scheme length. Combined with the new boundary and polar calculations, this gives [corollary 10.22](#) and [theorem 10.28](#); the local-term formula itself is classical, while its application to this ray fiber closes the zero-section gap. [Proposition 10.23](#) shows that this input is not replaceable by the previously known tame character equations plus positivity. Finally, BAS_1 proves only the small ray column [\(9.2\)](#); it does not prove the separate parity column [\(9.3\)](#).

11. A FINITE SANITY CHECK

The stable and unstable pieces can each be much larger than the final answer. Direct enumeration over prime fields gives the following values for $h = 1$:

q	d	total trace	stable trace	unstable trace
3	2	6	6	0
3	3	12	42	−30
3	4	24	114	−90
5	2	20	20	0
5	3	90	180	−90
5	4	400	700	−300

Here “stable” means $\deg \text{lcm}(D, E) \leq d$. The table is only a check of the exact decomposition; it also shows why discarding the unstable complex cannot be justified by dimension language alone. The source distribution includes the dependency-free verifier `tools/verify_prime_bar_small_cases.py`.

12. LOGICAL STATUS AND NEXT CONSTRUCTIONS

Statement	Status	Source of content
Poset Möbius number as reduced Euler characteristic	classical	Rota; topology of order complexes
Bar compression to the sign local system	existing theorem, specialized	Das–Howe, Proposition 6.3.2
The prime complex $\mathcal{P}_d$ with trace Λ	alternative split model	Möbius convolution; Sawin gives the alternating factorization identity
Degree-weighted Koszul enhancement and exterior comparison	proved globally after the stated choice of normalization	theorem 4.5
Euler-tail realization by the genuine filtered divisor bar	proved over $\mathbb{Q}_\ell$	Das–Howe, Theorem 6.5.4; theorem 4.11
Relative two-form trace identity	standard formalism	six functors and Grothendieck–Lefschetz
Affine-bundle range and singular-series extraction	standard calculation, repackaged	Chinese remainder theorem; Sawin–Shusterman, Proposition 6.2
Power saving for the residual trace	known arithmetically for large q	Sawin–Shusterman, Theorem 6.3
Marked rank-interval dg connection	proved globally over $\overline{\mathbb{Q}_\ell}$	theorem 4.18 ; filtered Das–Howe antisymmetrization
Resultant collision triangle	proved; no cancellation estimate	proposition 4.21
Cubical bar–Heath–Brown resolution	chain-level cube; standard vertices only in K_0	proposition 4.23
Marked cubical Type I/II comparison	open target	research Target 4.25
Residual weight–Betti theorem	open target	would be the new algebraic engine
Ray Fourier dictionary and RH baseline	classical	character orthogonality; Deligne; proposition 10.3
Nonzero-frequency interior cutoff	proved here, characteristic-free	theorem 10.5
Ordered ray-boundary cleanliness	proved for squarefree D_{∞^2}	theorem 10.9 and corollary 10.10
Fixed-frequency perversity and two-degree concentration	proved here, arbitrary characteristic	Gabber–Loeser; theorem 10.12 and corollary 10.14
Fixed-frequency top polar $P_{n,k}$	proved here, arbitrary characteristic	Sawin’s cycles; theorem 10.17
Generic rank $G_{n,k}$ and the point-wise ray bound	proved here, arbitrary characteristic	SGA 5 local terms; lemma 10.19 , theorem 10.21 , and corollary 10.22
BAS ₁ beyond 1/2 for $q > 266$	proved here	theorem 10.28 and corollary 10.29
Degree-one exact Kloosterman identity	classical transform, specialized here	theorem 10.24
Goldbach parity ray column	open target	(9.3)

The next constructions should be attempted in this order.

- (1) Prove the coherent cubical comparison in [research Target 4.25](#). The two ingredients are now separate theorems: the marked connection is [theorem 4.18](#), and the chain-level Čech cube is [proposition 4.23](#).
- (2) First recover the actual small-divisor/complement two-column split of [23, Corollary 6.1]. Independently localize the inner cubical resolution into Type I/II degree boxes. Only after the comparison maps commute should an outer middle-degree slab be called a bilinear E_1 -object; the outer large-divisor slab remains a separate complement/parity column.
- (3) Insert [theorem 10.28](#) into the full Goldbach column: combine it with the $\alpha = 0$ squarefree-progression theorem, perform Fourier inversion, and audit the finitely many coprimality and leading-coefficient strata in (9.2). No wild fixed-cycle estimate remains at this stage.

- (4) Attack the parity column (9.3). This is the step that would convert the exact Goldbach object into a new arithmetic theorem.
- (5) Only after these comparisons are available, retain the full S_d -representation or secondary Frobenius eigenvalues. Those refined invariants are where the algebraic formulation can carry information invisible in a bare big- O estimate.

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